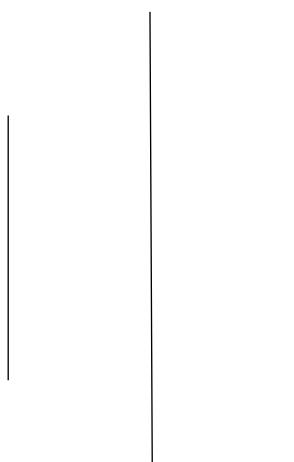




POKHARA UNIVERSITY
NEPAL COLLEGE OF INFORMATION TECHNOLOGY
BALKUMARI, LALITPUR



A

REPORT ON
SEISMIC ANALYSIS AND DESIGN OF
A RESIDENTIAL BUILDING

*A report submitted to the department of civil engineering in partial fulfillment
of the requirement for the degree of bachelor of civil engineering*

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ABSTRACT

Pokhara University offers a four-year Bachelor's Degree program in Civil Engineering, in which students are required to undertake a project work in the final semester to apply theoretical knowledge to practical engineering problems. This project work has been carried out to fulfill that academic requirement.

The selected project is titled "SEISMIC ANALYSIS AND DESIGN OF A RESIDENTIAL BUILDING." The main objective of this project is to gain adequate knowledge and practical understanding necessary for the analysis and design of structures.

This project focuses on the structural analysis and design of a residential building. It includes all stages of structural design such as determination of loads, preliminary design of structural members, structural analysis, and detailed design.

All loads acting on the building have been calculated in accordance with the relevant IS Codes and have been applied appropriately. The preliminary design involves the assessment of suitable dimensions for structural components such as slabs, beams, and columns, based on basic building parameters and codal provisions.

Structural analysis involves the evaluation of internal forces developed in structural members due to various types of loading, including dead load, live load, earthquake load, and other dynamic loads. Response spectrum analysis has been carried out to analyze the seismic behavior of the structure. Structural design includes proportioning and detailing of members to safely resist the induced forces throughout the service life of the structure. The design has been checked in compliance with seismic detailing requirements specified in the relevant codes to ensure earthquake resistance.

This report is the result of continuous effort, dedication, and perseverance. Any suggestions, comments, or constructive criticism for the improvement of this project will be highly appreciated.

ACKNOWLEDGEMENT

Nepal is one of the earthquake prone zones and at the same time it is one of the developing countries of the world. Different facilities are centered at major cities so construction of building is focused on city. The compulsion of construction of building high, grabs the attention of the structure engineers to be very crucial in terms of most efficient, effective and economic structural design of the overall building construction activity. In this connection, our team efforts have been focused to do project work on "**Seismic Analysis and Design of a Residential building**".

This entire project would probably have not been completed in this form on line without the invaluable guidance of **Assistant Prof. Mr. Lalit Bhatt**, our Project Advisor. We would like to thank him for his unlimited guidance, support and for mentoring the overall processes.

Finally, we gratefully acknowledge all the direct and indirect help, support and suggestion obtained to conclude this report.

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ABBREVIATIONS

A_g : Gross area of the section

A_c : Net area of concrete

A_{st} : Area of tensile steel in flexural member

A_{sc} : Area of compression steel

A_{sv} : Area of vertical stirrups

BM: Bending moment

b : Width of the beam

d : Effective depth of beam or slab

d' : Effective cover

D : Overall depth of beam or slab

DL: Dead load

E : Young's modulus of elasticity of steel

e_{min} : Minimum eccentricity

e_x, e_y : Eccentricity along X and Y axes respectively

EQ: Earthquake load

EQX1: Earthquake load along X-direction with dead and live load

EQX2: Earthquake load along X-direction with dead load only

EQZ1: Earthquake load along Z-direction with dead and live load

EQZ2: Earthquake load along Z-direction with dead load only

f_{ck} : Characteristic compressive strength of concrete

f_{sc} : Design stress in compression steel at the level of centroid of compression steel

f_{cc} : Design stress in concrete at the level of centroid of compression steel

f_y : Characteristic yield strength of steel

F_i : Design earthquake load on each column

h : Height of the column

l_d : Development length of the bar

l_{eff} : Effective length of the beam, slab, or column

L_x, L_y : Clear length of column or clear span of beam

LL: Live load

LLD: Least lateral dimension

M_u : Factored design moment

M_{ulim} : Limiting factored moment of resistance

M_{uy} : Factored design moment about Y-axis

M_{uy1} : Maximum moment capacity for bending along Y-axis

M_{uz} : Factored design moment about Z-axis

M_{uz1} : Maximum moment capacity for bending along Z-axis

p: Percentage of steel in the section

P_u : Factored design axial load

P_{uz} : Capacity of the cross section under pure axial load

Q_i : Design earthquake load on each floor level

RLL: Reduced Live Load

V_u : Factored shear force

VB: Total base shear

V_{us} : Factored strength of shear reinforcement

x: Spacing of the stirrups

α_n : An exponent that depends on the dimension of the cross section, the amount of reinforcement, concrete strength and yield stress of steel

τ_{bd} : Design bond stress

τ_c : Shear stress of the concrete

$\tau_{c,max}$: Maximum shear stress of the concrete

τ_v : Nominal shear stress

ϕ : Diameter of the steel bar

Table of Content

Contents

ABSTRACT	2
ACKNOWLEDGEMENT.....	3
ABBREVIATIONS	4
Table of Content	6
CHAPTER 1 : INTRODUCTION.....	8
1.1 BACKGROUND.....	8
1.2 Theme of Project.....	10
1.3 Objectives	10
1.4 Scope.....	10
1.5 Limitations:	10
1.6 Salient features of the project :.....	11
CHAPTER 2: REVIEW OF LITERATURE.....	12
Nepal National Building Code (NBC)	12
Indian Standard (IS) Codes	12
2.1 Loads	13
2.1.1 Vertical Loads	13
2.2 Load Combination.....	14
2.3 Structural Systems Of Buildings	14
2.4 Analysis	15
2.4.1 Analysis Using Standard software.....	15
ETABS	15
2.5 Design Philosophy.....	16
The Limit State Method:	16
2.6 Design Of Structural Member	19
CHAPTER 3: METHODOLOGY	19
3.1 Data Collection.....	20
3.2 Load Calculation	20
3.2.1 Vertical Load.....	20
3.3 Fundamental Time Period of Building (T).....	23
Elastic Seismic Coefficient	24
Reduction for Inelastic Behaviour.....	25
3.4.1. Preliminary design of Slab	26
3.4.2 Preliminary Design of Beam	28
3.4.3 Preliminary Design Of Column.....	30
CHAPTER 4 : STRUCTURAL ANALYSIS	33
4.1 Building Model.....	33
4.2 Storey Drift.....	34

4.3 Storey Displacement.....	38
Tabulated Plot Coordinates	42
4.3 Check For Eccentricity:.....	43
.....	43
Torsional irregularity check	45
CHAPTER 5 : DETAIL DESIGN OF MAIN STRUCTURAL COMPONENTS	54
5.1 Design of slab.....	54
5.2 Design Of Column.....	68
5.3 Design of Staircase	80
.....	81
5.4 Design of Basement wall.....	86
5.5 DESIGN OF MAT FOOTING	94
Axial Force, Shear Force and Bending Moment Diagram	100

CHAPTER 1 : INTRODUCTION

1.1 BACKGROUND

Nepal is located in a highly seismically active region due to the ongoing convergence of the Indian and Eurasian tectonic plates. This tectonic interaction results in frequent earthquakes that pose a significant risk to buildings and infrastructure. The Kathmandu Valley is particularly vulnerable to seismic hazards because of its proximity to active fault systems and the presence of deep alluvial soil deposits, which are known to amplify seismic ground motion. Past seismic events, most notably the 2015 Gorkha earthquake, have highlighted the susceptibility of residential reinforced concrete (RC) buildings to severe damage when seismic design provisions are inadequately addressed.

Residential RC buildings constitute a major portion of urban development in Kathmandu District, where mid-rise buildings with basements are increasingly common due to land constraints. Such buildings experience considerable seismic demand owing to increased mass, vertical and plan irregularities, and stiffness variation introduced by basement levels. Therefore, detailed seismic analysis and earthquake-resistant design are essential to ensure structural safety, serviceability, and acceptable performance during seismic events. The fundamental objective of seismic design is not to eliminate damage entirely, but to prevent collapse and ensure life safety through adequate strength, stiffness, and ductility.

Seismic analysis and design practices in Nepal are governed by the Nepal National Building Code (NBC). NBC 105:2020 – Seismic Design of Buildings in Nepal provides the primary framework for seismic zoning, design response spectra, importance factors, load combinations, and analysis procedures, including equivalent static and dynamic analysis methods. In addition, NBC 205:2024 – Ready-to-Use Design (RUD) Guidelines for Reinforced Concrete Buildings offers simplified prescriptive design and detailing provisions intended mainly for low-rise and regular RC buildings, allowing rapid and conservative design without detailed analytical procedures. While NBC 205:2024 is useful as a supplementary reference, detailed seismic analysis is mandatory for mid-rise buildings located in high seismic zones such as Kathmandu.

Indian Standards are also widely referred to in Nepalese engineering practice to complement NBC provisions. IS 1893 (Part 1):2016 is used for the determination of seismic forces and structural response parameters, while IS 456:2000 governs the design of reinforced concrete members. Furthermore, IS 13920:2016 provides detailed requirements for ductile detailing of RC structures subjected to seismic forces, ensuring enhanced energy dissipation capacity and improved seismic performance.

The outcome of the seismic analysis forms the basis for the design of structural components

including slabs, beams, columns, and foundations, ensuring compliance with strength, serviceability, and ductility requirements prescribed by the governing codes. Through the application of national and international seismic design standards, this project aims to demonstrate a systematic and code-compliant approach to the seismic analysis and design of a mid-rise residential building in a high seismic zone, thereby contributing to the development of earthquake-resilient residential infrastructure in Nepal.

1.2 Theme of Project

This group under the project has under taken the Seismic Analysis And Design Of Residential Building. The main aim of the project work under the title is to acquire knowledge and skill with emphasis of practical application of various available codes of practices is another aim of the work.

1.3 Objectives

The main objectives of the project are analysis, design of a residential building. In brief, the objectives are as follows:

Preliminary design of the major components of building like beam, column, slab, staircase, foundation., shear walls.

Analysis of major components of building like beam, column, slab, staircase, foundation, shear walls VIZ. ETABS.

1.4 Scope

Since, Construction of earthquake proof buildings is almost impossible, so we focus our study on analysis and design of seismic resistant building. The project will be for a multistoried of 5-Storey with basement.

- Project work is carried out to structural analysis and design of a building.
- Design and detailing of typical elements of following structural members is performed:
 - ✓ Slab
 - ✓ Beam
 - ✓ Column
- The project does not take into account the existing soil condition of the locality and soft soil is assumed .
- Three-Dimensional analysis is done using software (ETABS).
- Limit state method is used for the design of structural components.

1.5 Limitations:

Due to various constructions prevailing in the course of the project work, the study is limited in following notable steps:

- Data manipulation is checked manually with underlying concepts but some similar sections are solely on software due to limitations.

1.6 Salient features of the project :

Project of apartment building with :

Table 1: Building Information

Project Area	Kathmandu
No. of Storey	5
Shape of Building	Rectangular
Height of Storey	2.870 m
Toatal Height	16.992 m
Length	9.779 m
Width	15.22 m
Types of Staircase	Openwell
Plinth Area	148.83 m
Types Of Structure	Frame Structure

CHAPTER 2: REVIEW OF LITERATURE

All engineering designs are founded on previously established knowledge, experience, and documented references. Even when a completely new design is introduced, its fundamental principles are closely related to existing theories, standards, and practices. For the completion of this project on the design of an earthquake-resistant multi-storeyed building, various codes, standards, and reference materials were studied under the guidance of the project supervisor.

Nepal National Building Code (NBC)

The Nepal National Building Code (NBC) has been developed to standardize building construction practices in a manner suitable for the Nepalese context. Since the development of the code is relatively recent, it still lacks several supporting documents. To address this limitation, the NBC frequently refers to relevant Indian Standards (IS Codes).

The NBC categorizes the level of design and construction sophistication into the following four types:

International state-of-the-art structures

Professionally engineered structures

Buildings of restricted size designed using simple rules of thumb

Remote rural buildings where strict control is impractical

The present project falls under the second category, i.e., professionally engineered structures. Furthermore, the NBC permits the use of international codes provided their provisions satisfy NBC requirements. Hence, the use of Indian Standard (IS) codes in this project is justified.

Indian Standard (IS) Codes

The following Indian Standard codes have been referred to in this project:

A) IS 456:2000 – Plain and Reinforced Concrete: Code of Practice

This code provides guidelines for the design of reinforced concrete structural elements such as beams, columns, slabs, staircases, and footings. It also specifies design criteria for limit states of flexure, shear, compression, and torsion, along with serviceability requirements such as deflection and cracking.

B) IS 875:1987 – Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures

This code specifies the procedures for estimating design loads acting on structures and is divided into five parts:

Part 1: Unit weights of construction materials and stored materials

Part 2: Imposed (live) loads for different types of buildings and usage conditions

Part 3: Wind loads (not applicable to the present design)

Part 4: Snow loads (not applicable to the present design)

Part 5: Loads due to temperature effects, earth pressure, hydrostatic pressure, and prescribed load combinations

C) IS 1893 (Part 1):2002 – Criteria for Earthquake Resistant Design of Structures

This code deals with the evaluation of seismic forces acting on structures and the design principles for earthquake-resistant buildings. It addresses the fundamentals of seismic engineering, including methods for determining seismic loads and the effects of structural irregularities on seismic response.

D) IS 4326:1993 (Reaffirmed 2003) – Earthquake Resistant Design and Construction of Buildings: Code of Practice

While IS 1893 focuses primarily on the analytical and mechanical aspects of earthquake engineering, IS 4326 provides practical guidelines for earthquake-resistant design and construction practices to improve the overall seismic performance of buildings.

2.1 Loads

Structurally speaking, buildings are built to support loads. There are different types, which one come across and have to be dealt with in then analysis and design of any structure.

2.1.1 Vertical Loads

2.1.1.1 Dead Load

The load, which is ever present and ever acting on building, is the dead load which consists of self-weight of members, finishes, plaster, false ceiling, parapets, balconies, fixed or permanent equipments etc. The load acting on a column is calculated from all the beams framing into it . Dead load should be calculated very accurately, ass it comprises most of the building loads.

NBC 105 , IS 875-1987 PART I gives a schedule of unit weights of building materials and

it is used extensively to calculate the dead load.

In addition, partition loads are also important to be considered. In multistory building, 100mm thick brick wall anywhere adds substantially to the load of the building and affects the design of slabs, beams, columns and footing too.

2.1.1.2 Live Load

Next in importance to dead load, is the live load, which is caused by the use of building. The magnitude of live load depends upon the type occupancy of the building. Live loads are generally high on the floors depending on the activity that are carried on them, while it is low on roof, which may or may not be accessible. The live load distribution varies with time.

Live loads should be calculated as per NBC 105.

2.2 Load Combination

Where seismic load effect is combined with other load effects, the following load combination shall be adopted.

- $1.2DL + 1.5 LL$
- $1.2DL + 1.5 LL + S$
- $DL + \lambda LL + S$
- $DL + \lambda LL \pm E$
- $0.9DL \pm E$

Where,

$S = 1.2S_{so}$ for snow load

$= 1.2 S_{lp}$ for static liquid pressure

$= 1.5 S_{ep}$ for earth pressure

$\lambda = 0.6$ for storage facilities

$= 0.3$ for other usage

2.3 Structural Systems Of Buildings

To achieve substantial economy in the structural design of a building, the proper selection of a structural system is more important than designing only the critical sections of structural members. A structural system suitable for a low-rise building may not be appropriate for a high-rise structure.

For buildings up to four storeys, a reinforced concrete framed structure is found to be economically efficient in resisting both vertical and horizontal loads. Therefore, in the present case, a reinforced concrete framed structural system is adopted for the building.

2.4 Analysis

The analysis of a building structure can be carried out using two major approaches:

- a. Manual method
- b. Using standard software

In the manual method of analysis, the plane frame approach is commonly adopted. The plane frame method includes the following techniques:

- i. Moment distribution method
- ii. Matrix method
- iii. Approximate method

For multi-storey buildings, the degree of static indeterminacy is very high. As a result, the moment distribution method and matrix method become tedious and time-consuming when compared to approximate methods of analysis. Hence, approximate methods are preferred for preliminary analysis and design of such structures.

The cantilever method is generally used for the analysis of buildings having less than 35 storeys and subjected to lateral loads such as wind or earthquake forces.

The following assumptions are made in the cantilever method:

- Points of contra-flexure in each structural member are assumed to occur at the mid-span of beams and at the mid-height of columns.
- The direct (axial) stresses developed in columns due to horizontal forces are assumed to be directly proportional to their distance from the centroidal vertical axis of the structural frame.

2.4.1 Analysis Using Standard software

Structural analysis can be carried out using standard software through the following approaches:

- a. Two-dimensional (2D) analysis
- b. Three-dimensional (3D) analysis

Among the various standard structural analysis software available, **ETABS** has been used for the purpose of this project.

ETABS

ETABS (Extended Three-Dimensional Analysis of Building Systems) is an advanced structural analysis and design software specifically developed for the analysis and design

of building structures. It provides an integrated environment for modeling, analysis, design, and detailing of reinforced concrete and steel structures.

ETABS is user-friendly and highly efficient for multi-storey building analysis. The software allows the creation of structural models using graphical interfaces, enabling accurate definition of geometry, material properties, load cases, and boundary conditions. ETABS performs analysis using advanced numerical techniques and provides reliable results in terms of forces, moments, deflections, and design outputs in accordance with relevant design codes.

Note: Only ETABS software has been used for the structural analysis in this project work.

2.5 Design Philosophy

There are three commonly adopted philosophies for the design of reinforced concrete structures, namely:

- a. Working Stress Method
- b. Ultimate Load Method
- c. Limit State Method

Among the above three methods, the **Limit State Method** has been adopted for the design of the structural elements in this project.

The Limit State Method:

The objective of design based on the limit state concept is to ensure that the structure performs satisfactorily throughout its intended life span with an acceptable level of safety and serviceability. The structure should not reach a condition beyond which it becomes unsafe or unserviceable during its design life.

All relevant limit states must be considered during design to ensure adequate safety and serviceability. The most important limit states that are examined in the design process include:

- Limit state of collapse
- Limit state of serviceability

a. Limit State of Collapse

The limit state of collapse corresponds to the maximum load-carrying capacity of a structure. Violation of this limit state indicates structural failure in the sense that a clearly defined limit of structural usefulness has been exceeded. However, it does not necessarily imply total collapse of the structure.

This limit state is associated with failure due to:

- Flexure
- Compression
- Shear
- Torsion

b. Limit State of Serviceability

In addition to satisfying the limit state of collapse, reinforced concrete structures must also meet the requirements of serviceability. The limit state of serviceability relates to conditions under which excessive deformation or other effects may impair the normal use and functioning of the structure or its components.

This limit state is concerned with checking members where excessive deformations may restrict usability, comfort, or durability. The limit state of serviceability may correspond to:

- Excessive deflection
- Cracking
- Vibration

Serviceability and Ultimate Limit State Considerations in Seismic Design

In structural engineering design, particularly for buildings subjected to seismic actions, the performance of structural elements is evaluated using two principal criteria: **the Serviceability Limit State (SLS)** and **the Ultimate Limit State (ULS)**. These limit states ensure that the structure remains both safe against collapse and functional during its service life.

Ultimate Limit State (ULS)

The Ultimate Limit State represents the condition beyond which the structure or structural element fails to satisfy the requirements for safety. Under seismic loading, ULS ensures that the building possesses sufficient strength, stability, and ductility to resist collapse during severe earthquake events.

ULS verification involves checking:

- Structural strength of beams, columns, and slabs
- Stability against overturning and buckling

- Load-carrying capacity under factored loads
- Formation of plastic hinges and energy dissipation capacity

In seismic design, load combinations include amplified earthquake forces and reduced material strengths to represent extreme conditions. Satisfaction of ULS guarantees life safety by preventing structural collapse.

Serviceability Limit State (SLS)

The Serviceability Limit State corresponds to conditions under which the structure remains functional, comfortable, and visually acceptable during normal usage and moderate seismic events. While ULS addresses structural safety, SLS focuses on performance and usability.

SLS verification involves checking:

- Excessive lateral displacement (inter-storey drift)
- Cracking in reinforced concrete elements
- Vibrations affecting occupant comfort
- Permanent deformation affecting structural integrity
- Damage that may impair building functionality

Under seismic analysis, SLS ensures that earthquake-induced deformations remain within permissible limits so that non-structural components and building services are not adversely affected.

Role in Seismic Design

- For residential buildings subjected to earthquake loading: ULS ensures structural survival during major earthquakes.
- SLS ensures acceptable performance during frequent or moderate earthquakes.
- Both limit states must be satisfied to achieve a design that is structurally safe, durable, and suitable for occupancy.

Summary:

Table 2: Summary difference of ULS & SLS

Aspect	ULS	SLS
Goal	Prevent collapse	Ensure usability
Load level	Extreme	Normal
Concern	Strength & stability	Deflection, cracking, vibration
Consequence of failure	Dangerous	Uncomfortable / unfit

2.6 Design Of Structural Member

The design of reinforced concrete structural elements such as slabs, beams, columns, and footings is carried out in accordance with the limit state method. All critical sections of reinforced concrete members are designed for the forces acting at those sections, as obtained from the structural analysis of the members.

CHAPTER 3: METHODOLOGY

The methodology adopted for the analysis and design of the five-storey apartment

building is explained in the following sections.

3.1 Data Collection

All the data required for this project were collected from different standard literature sources. The design parameters were taken from the **NEPAL BUILDING CODE and recommended IS code**.

3.2 Load Calculation

The load calculations were carried out based on NBC 105. The unit weights of construction materials used in the building were taken from the relevant codes. The thickness of materials was selected according to design requirements.

3.2.1 Vertical Load

The loads acting on slabs are classified as follows:

- i. **Dead Load** – Self-weight of the slab
- ii. **Live Load** – Loads taken from the relevant design codes

The loads acting on beams include:

- a. Self-weight of the beam
- b. Load transferred from the slab
- c. Load due to walls
- d. Load transferred from the staircase

The load on the columns was calculated by summing the total dead load and live load acting on them according to their tributary area, and then adding the self-weight of the column.

a. Self-Weight of the Beam

After determining the preliminary beam size through manual calculations, the same section dimensions were assigned to the beam elements in **ETABS**. Once the beam section is defined, **ETABS automatically calculates the self-weight of the beam** based on the material density and cross-sectional properties.

The self-weight is considered as a **uniformly distributed load (UDL)** acting along the span of the beam. This load is internally computed by ETABS using the unit weight of reinforced concrete, typically taken as **25 kN/m³**, and the cross-sectional area of the beam.

For verification purposes, the self-weight of the beam can also be calculated manually using

the expression:

$$\text{Self-weight per meter length} = b \times D \times \gamma_c$$

where:

- b = width of the beam (m)
- D = overall depth of the beam (m)
- γ_c = unit weight of reinforced concrete ($\approx 25 \text{ kN/m}^3$)

The manually calculated self-weight closely matches the value automatically generated by ETABS, confirming the correctness of the assigned beam section and load modeling.

b. Load Transfer from Slab to Beam

The load from the slab, including both dead load and live load, can be transferred to the beam as an equivalent **uniformly distributed load** by two methods:

- **Triangular Load Distribution**
- **Trapezoidal Load Distribution**

For a **two-way slab**, load distribution is done according to the **yield line theory**. The loads acting on beams supporting a two-way slab and carrying uniformly distributed load are calculated in accordance with **Figure 1** shown below.

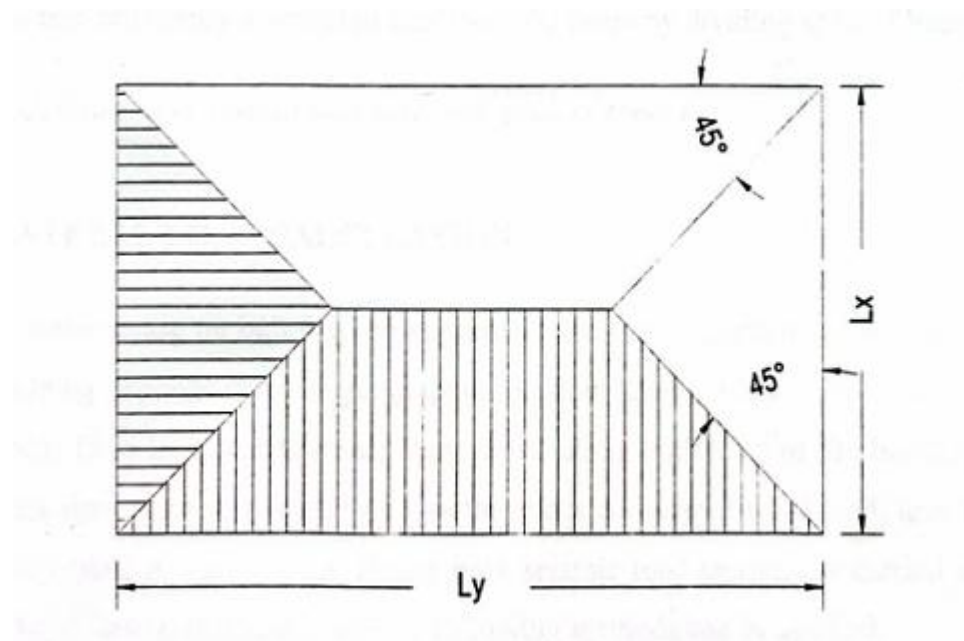


Figure 1: Load transfer from Slab to Beam

The slab is supported by four beams.

The **longer beams** carry **trapezoidal loads**, while the **shorter beams** carry **triangular loads**.

The following formulas are used to convert the **plan trapezoidal and triangular loads** into **line loads** acting on the beams.

Triangular and Trapezoidal Load

$$\text{Triangular Load} = \frac{w \times l_x}{3}$$
$$\text{Trapezoidal Load} = \frac{w \times l_x \times r}{r + 2}$$

Where:

$$r = \frac{l_y}{l_x}$$

w = intensity of load in kN/m²

c. Load Due to Wall

The load from masonry walls, including self-weight of the wall and any plaster or finishing loads, is transferred directly to the supporting beams as a **uniformly distributed line load (UDL)** along the length of the beam. Unlike slab loads, wall loads act vertically and are assumed to be uniformly distributed over the beam span supporting the wall.

For walls resting centrally on beams, the entire wall load is transferred to the beam below. In the case of walls resting on slabs, the wall load is first transferred to the slab and then distributed to the supporting beams along with the slab loads.

The wall load per unit length acting on the beam is calculated using the following expression:

$$\text{Wall Load (kN/m)} = t \times h \times \gamma_m$$

Where:

- t = thickness of wall (m)
- h = height of wall including parapet, if any (m)
- γ_m = unit weight of masonry (kN/m³)

Additional loads due to plaster on both faces of the wall may be added as:

$$\text{Plaster Load (kN/m)} = 2 \times t_p \times h \times \gamma_p$$

Where:

- t_p = thickness of plaster (m)
- γ_p = unit weight of plaster (kN/m³)

The total wall load is then applied as a **uniformly distributed load** on the supporting beam in the structural analysis and design.

d. Load Transfer from Staircase

The reactions produced by the staircase are first calculated. These reactions are then converted into **uniformly distributed loads (UDL)** on the supporting beams by dividing them by the span of the beam.

3.3 Fundamental Time Period of Building (T)

The **fundamental time period (T)** of a building is the time required to complete one cycle of free vibration in its fundamental mode. It is an important dynamic property of a structure and plays a key role in seismic analysis because the design seismic force depends directly on the natural period of vibration.

For reinforced concrete moment-resisting frame (RC MRF) buildings without infill walls, the empirical formula for estimating the fundamental time period is:

$$T = 0.0075 \cdot H^{0.75}$$

where:

- **T** = Fundamental time period (seconds)
- **h** = Height of the building in meters (measured from base to top of structure)

This formula is provided in seismic design codes (Clause 7.3.2 / relevant seismic code provision) for preliminary estimation when detailed dynamic analysis is not performed.

Basis of the Formula

- The expression is **empirical**, derived from observations and vibration studies of actual buildings.
- It relates the time period primarily to the **height of the building**, since taller buildings are more flexible and therefore have longer time periods.
- The exponent **0.75** reflects the nonlinear relationship between height and flexibility.
- The coefficient **0.075** depends on the structural system (RC moment-resisting frame in this case).

Importance in Seismic Design

The fundamental time period is used to:

1. Determine the **design horizontal seismic coefficient** from the response spectrum.
2. Calculate the **design base shear**.
3. Assess dynamic characteristics of the structure.

A higher time period generally results in lower seismic acceleration demand (depending on the response spectrum), but excessive flexibility may cause serviceability issues such as large lateral displacements.

$$\begin{aligned}\text{Time Period}(T) &= 0.0075 \cdot H^{0.75} && [\text{cl.7.3.2}] \\ &= 0.075 \cdot 17.221^{0.75} \\ &= 0.634\end{aligned}$$

1. Base Shear Coefficient (C)

The **base shear coefficient (C)** represents the ratio of total design horizontal seismic force (base shear) to the seismic weight of the structure. It is used to calculate the **design base shear (V)** as:

$$V = C \times W$$

where:

- **V** = Design base shear
- **C** = Base shear coefficient
- **W** = Seismic weight of the building

According to **NBC 105**, the base shear coefficient depends on the seismic hazard, structural ductility, importance of the building, soil condition, and structural system.

Base Shear Coefficient for Ultimate Limit State Method:

For Ultimate Limit State, NBC 105 provides:

$$C = C(T_i) R_u \Omega$$

where:

- **C(T_i)** = Elastic seismic coefficient at fundamental period T
- **R_u** = Ductility factor
- **Ω** = Overstrength factor

Elastic Seismic Coefficient

$$C(T) = C_h(T) \times Z \times I$$

where:

- **C_h(T)** = Spectral shape factor (depends on soil type and time period)
- **Z** = Seismic zone factor
- **I** = Importance factor

Reduction for Inelastic Behaviour

Buildings are designed to undergo controlled inelastic deformation during strong earthquakes. Therefore, the elastic force is reduced using:

- i. Ductility Factor (R_u)
- ii. Overstrength Factor (Ω)

For Ultimate Limit State, $C = \frac{C(T_i)}{R_u \cdot \Omega}$ [cl.6.1.1]

Here, $C(T_i) = Ch(T) \cdot Z \cdot I$

Now, $Ch(T) = \alpha$ [cl.4.1.2]

$= 2.25$

$Z = 0.35$ (for soil type D, Kathmandu) [cl.4.1.4]

$I = 1$ [cl.4.1.5]

So, $C(T_i) = 2.25 \cdot 0.35 \cdot 1$
 $= 0.7875$

Ductility Factor (R_u) = 4 [cl.5.4.2 Table 5-2]

Overstrength Factor (Ω) = 1.5 [cl.5.4.2 Table 5-2]

So,

Base Shear Coefficient for Ultimate Limit State (C) = $\frac{0.7875}{4 \cdot 1.5} = 0.13125$

Base Shear Coefficient for Serviceability Limit State Method:

Similarly For Serviceability Limit State, $C = \frac{C_s(T_i)}{\Omega}$

Here, $C_s(T_i) = 0.20 \cdot C(T)$ [cl.4.2]

$= 0.20 \cdot 0.7875$

$= 0.1575$ [cl.5.4]

$\Omega = 1.25$

So, Base Shear Coefficient for Serviceability Limit State (C) = $\frac{0.1575}{1.25} = 0.126$

2. Building Height exponent (k)

From NBC 105:2025, By Interpolation using the value of C , for both Limit State Method and Serviceability Limit State Method:

Building Ht exponent (k) = 1.0687

3.4 Preliminary Design

The approximate size of the structural elements were determined through the preliminary design so that after analysis the pre-assumed dimensions might not deviate considerably, thus making the final design for safe purpose. Approximates size of various elements have been determined as follows:

3.4.1. Preliminary design of Slab

We consider Slab DC-23 as Critical Slab :

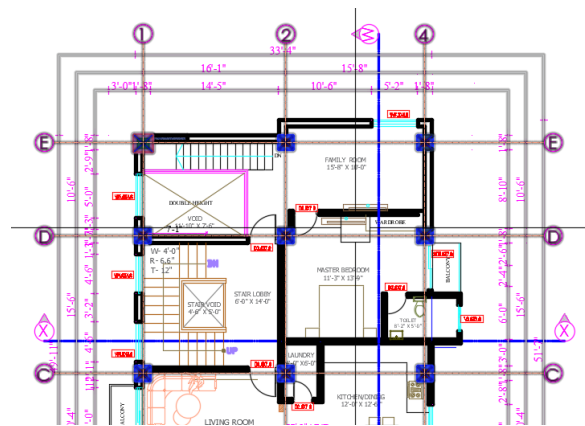


Figure 2: Preliminary design of column

Using the deflection control criteria defined in IS456:2000, Clause 23.2

$$L/d_{\min} = 26 * \text{Modification Factor}$$

For modification factor,

$$F_s = 0.58 * f_y * \frac{\text{Area of steel required}}{\text{Area of steel provided}}$$

Assuming area of steel required tends to or equal to area or steel provided,

$$F_s = 0.58 * 500 * 1 = 290 \text{ N/mm}^2$$

Assuming 0.12% of steel for tension reinforcement.

So, Modification Factor = 1.6 (Clause 23.2.1 (c) fig no. (4), IS 456: 2000)

Therefore, $L/d = 26 * 1.6$

Effective depth (d) = 113.58 take 130mm

Adopt Overall Depth (D) = $130 + 10/2 + 15 = 150\text{mm}$

(Here for minimum Clear Cover = 15mm) , IS 456: 2000

Check :

Self Weight = $25 \text{ KN/m}^3 * 0.150 \text{ m} = 3.75 \text{ KN/m}^2$

Imposed Load = 3 KN/m^2

Floor finish Load = 1 KN/m^2

Total Load = 7.75 KN/m^2

Adopt B = 1000mm

Factored Design Load (Wu) = 11.625 KN/m

Then,

$\mu = (wl^2/8) = 32.442 \text{ KNm}$

$d = \sqrt{\frac{\mu}{0.133f_{ck} B}} = 98.77 \text{ mm}$, take $99 \text{ mm} < 150 \text{ mm}$ OK

Therefore adopt Overall Depth (D) = 150 mm.

3.4.2 Preliminary Design of Beam

We Consider Beam (1-2,B-B) as critical beam.

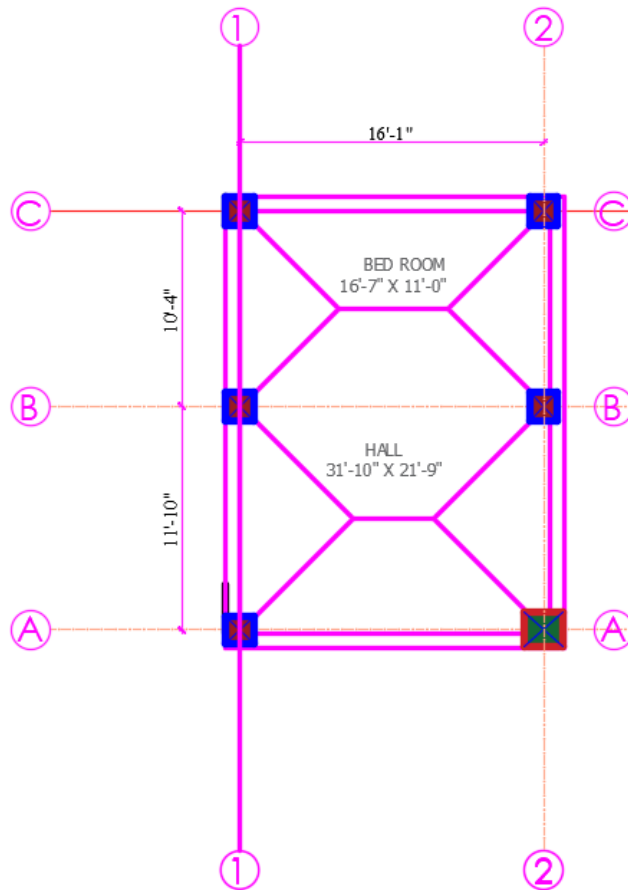


Figure 3: Preliminary Design of Beam

$$L_y = 16'-1'' = 4.902\text{m},$$

$$L_x = 11'-10'' = 3.6068\text{m},$$

$$\text{Effective span(lef)} = 4.902\text{m}$$

$$\text{Grade of steel} = f_e 500$$

$$\text{Unit weight of RCC} = 25\text{KN/M}^3$$

$$\text{Support width(bs)} = 230\text{mm}$$

$$\text{Partial safety factor (DL+LL)} = 1.5$$

$$\text{Assume, } l/d = 15$$

$$d = (4.902 \times 1000 / 15) = 326.8\text{mm}$$

Adopt effective depth (d)= 350mm which is >bs

IS 456 Table 16 says Clear Cover =20 for mild cover and 30 for moderate condition

$$d' = 25+8+16/2 = 44 = 50$$

Assume effective cover(d')=50mm

Overall depth of beam (D) =400mm

Assume d/b =1.5

Width(b)=233.336mm

Adopt B = 250mm

Hence The Dimension of Beam is 250 mm* 400 mm

3.4.3 Preliminary Design Of Column

The design of column is governed by limit state of axial compression and bending moments about two axes. Shear in column is small and shear stress work out to be safe. Stirrups in columns are provided mainly for holding column bars in place and making them strong bulking and bursting as these bars come under direct compression. Moments in column change sign in each storey, so that , we generally provide symmetrical bar arrangement in a column section and steel area is kept constant through out a given storey.

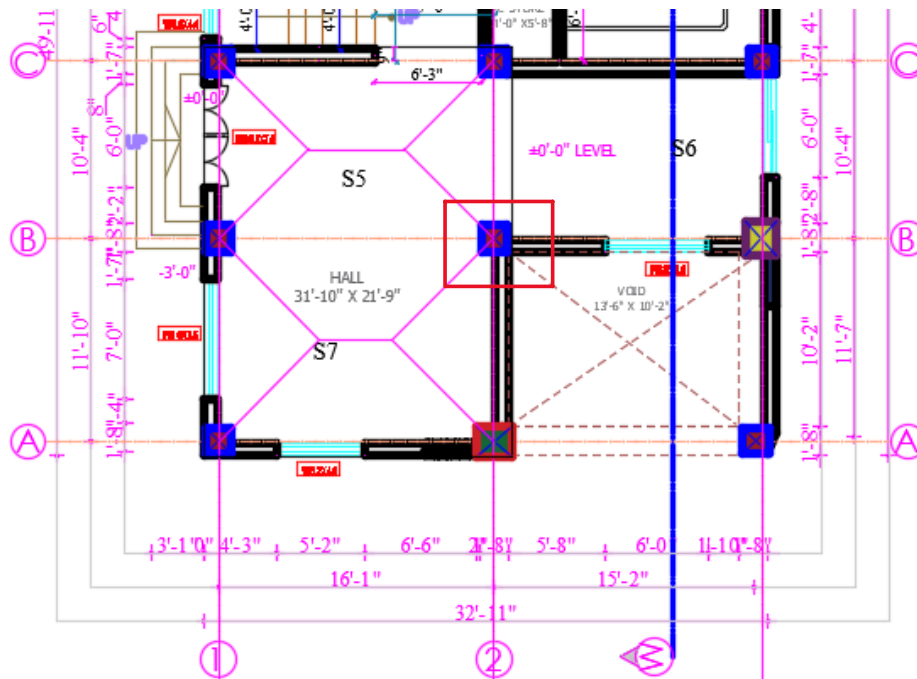


Figure 4: Preliminary Design of column

We consider B-B, 2-2 as Critical column
for load from slab to column

$$= 4.4387 \times 3.3782$$

$$= 16.346 \text{ m}^2$$

$$\text{Self weight of slab} = (25 \text{ kN/m}^3 \times 0.150 \text{ m}) \times 16.346 \text{ m}^2$$

$$= 61.2975 \text{ kN}$$

$$\text{Live load} = 2 \text{ kN/m}^2 \times 16.346 \text{ m}^2$$

$$= 32.692 \text{ kN}$$

$$\text{Floor finish load} = 1 \text{ kN/m}^2 \times 16.346 \text{ m}^2$$

$$= 16.346 \text{ kN}$$

Since we are designing column of basement
the column carries 6 slabs.

Now,

$$6 \times (\text{total factored load})$$

$$6 \times (1.5 (61.2975 + 32.692 + 16.346))$$

$$= 993.0195 \text{ kN}$$

Load from beam

Load from beam (B-B)

$$\text{Length} = 2451.1 + 2387.6$$

$$= 4838.7 \text{ mm}$$

$$25 \times b \times D \times L = 25 \text{ kN/m} \times 0.250 \times 0.400 \times 4.8263$$

$$= 12.056 \text{ kN}$$

Load from beam (2-2)

$$\text{Length} = 1574.8 + 1803.4$$

$$= 5378.2$$

$$25 \times b \times D \times L = 25 \text{ kN/m} \times 0.250 \times 0.400 \times 3.3782$$

$$= 12.096 \text{ kN}$$

Since we have 6 storey including base

(It carries slab)

$$= 6 \times (1.5 \times (12.056 + 8.4455))$$

$$= 184.87 \text{ kN}$$

Load from wall

= $Y_{\text{wall}} \times \text{thickness of wall} \times \text{height of wall} + (Y_{\text{plaster}} \times \text{thickness of plaster on both side} \times \text{height})$

$$= 19 \times 0.230 \times 2.7432 +$$

$$20 \times 0.012 \times 2 \times 2.7432$$

$$= 13.304 \text{ kN/m}$$

$$\text{for } 4.4387 \text{ m wall} = 64.374$$

$$\text{for } 3.3782 \text{ m wall} = 44.943$$

(thickness of plaster according to NBC 105)

12 mm internal

15 mm external

ceiling = 6

Total load from wall

$$= 59.374 + 44.943$$

$$= 109.317$$

Factored

$$= 1.5 \times 109.317$$

$$= 163.975 \text{ kN}$$

Since the column of basement carries 5 no of storey's wall

$$= 5 \times 163.975$$

$$= 819.877 \text{ kN}$$

Load of the column's

Assume a trial column of 300 mm x 300 mm

$$\text{Self wt} = 25 \times 0.300 \times 0.300 \times 2.8702$$

$$= 6.45795$$

$$6 \text{ no. of column} = 6 \times 6.45795$$

$$= 38.7477$$

$$\text{Factored} = 1.5 \times 38.744$$

$$= 58.125$$

Total weight acting on critical column (intermediate)
 = load from column + beam + wall + column's self wt
 = $993.0195 + 184.87 + 763.975 + 58.125$
 = 1399.986 kN

Using M25 grade and Fe500 steel
 Assuming short axially loaded column
 $P_u = 0.4 f_{ck} A_c + 0.67 f_y A_{sc}$

Assuming 1.5% of steel
 Using cl: 26.5.3.1
 $P_n = 18$

$$1399.986 = 0.4 \times 25 \times (100 - 1.5)/100 A_g + 0.67 \times 500 \times (1.5/100) A_g$$

$$A_g = 138595.32 \text{ mm}^2$$

Assuming square columns
 $D = B = \sqrt{138595.32}$
 = 372.22 mm
 Adopt column of size
 400 mm x 400 mm

Calculation of reinforcement

$$A_{sc \text{ req}} = 1.5/100 \times (400 \times 400)$$

$$= 2400 \text{ mm}^2$$

Provide 16 mm bar
 No. of bars = $2400 / (\pi \times 16^2 / 4)$
 = 11.93
 = 12 nos.

$$A_{st \text{ provided}} = 12 \times \pi \times 16^2 / 4$$

$$= 2412.74 > A_{st \text{ reqd}}$$

OK

Design of lateral ties

cl: 16.5.3.2.1 PN-7

Pitch of lateral ties shall be least of

(i) 16Φ
 = 16×16
 = 256 mm

(ii) least lateral dimension
 = 400 mm

(iii) 300 mm

Adopt pitch = 256 mm

Dia of lateral ties

$$= 1/4 \times 16$$

$$= 4 \text{ mm} (< 6 \text{ mm})$$

Adopt 7 mm

CHAPTER 4 : STRUCTURAL ANALYSIS

4.1 Building Model

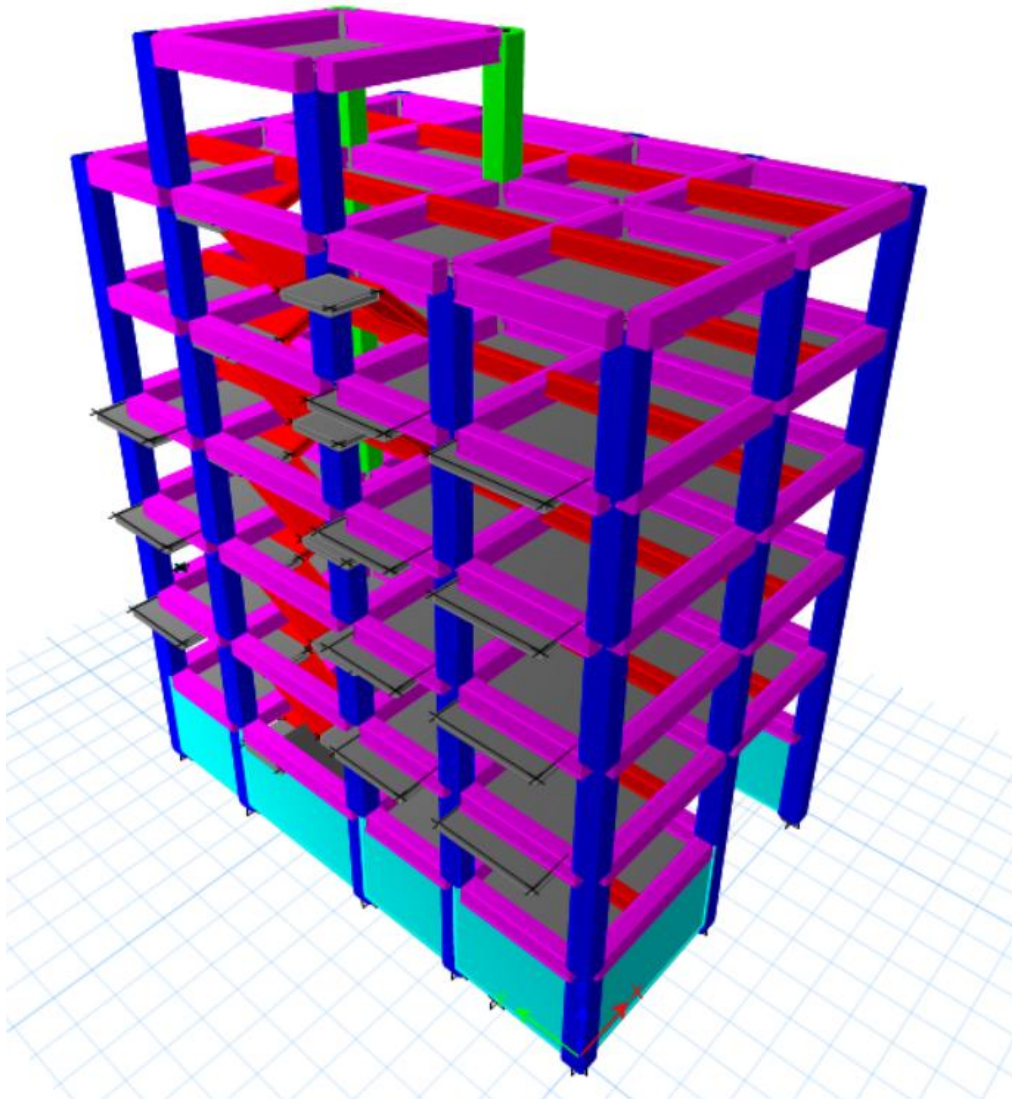


Figure 5 : South-West Elevation

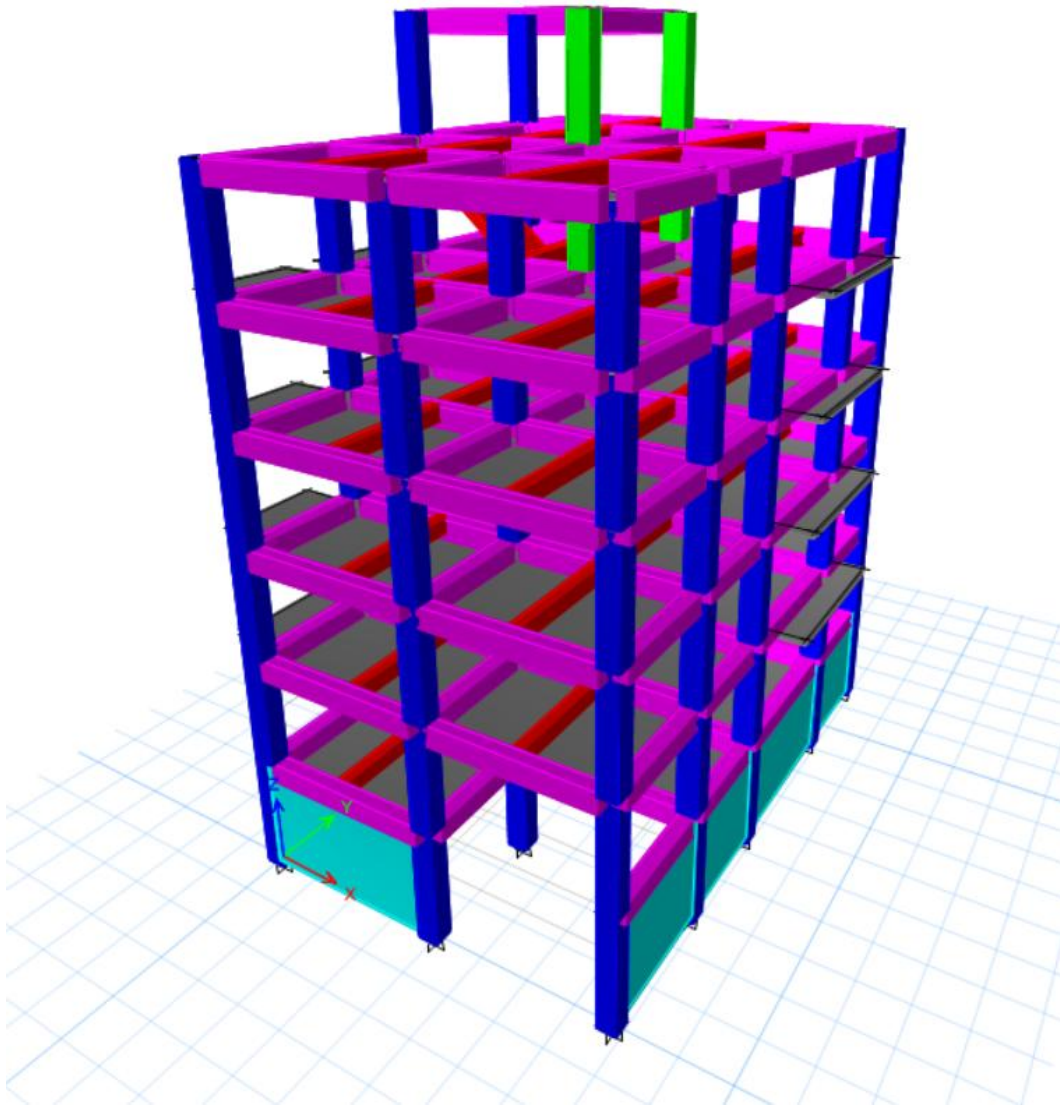


Figure 6: South- East Elevation

4.2 Storey Drift

Storey drift is the relative displacement between the floors above and/or below the storey under consideration. As per NBC 105: 2025 story drift in any story shall not exceed 0.006 times the storey height. The limitation on storey drift is necessary to avoid discomfort to occupants of the building and to save non-structural elements from damage. ETABS analysis directly generates the drift in form ratio so that the result can be directly compared with the permissible drift ratio of 0.006.

Story Response - Maximum Story Drifts

Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp1		
Display Type	Max story drifts	Story Range	All Stories
Load Case	EQx SLS	Top Story	Story7
Output Type	MaxMin	Bottom Story	Base

Plot

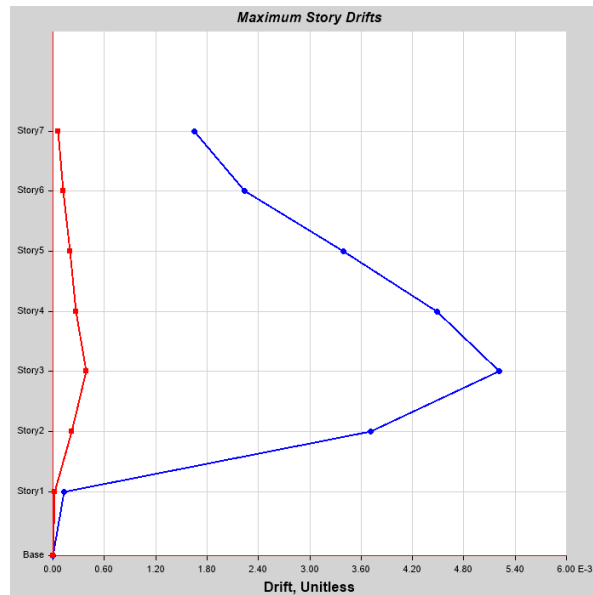


Figure 7 : Graph of maximum storey drift under EQx SLS

Tabulated Plot Coordinates

Story Response Values

Table 3 : Storey response values of maximum storey drift under EQx SLS

Story	Elevation	Location	X-Dir	Y-Dir
	m			
Story7	20.2692	Top	0.001653	0.000061
Story6	17.399	Top	0.002235	0.00012
Story5	14.5288	Top	0.003394	0.000195
Story4	11.6586	Top	0.00449	0.000269
Story3	8.7884	Top	0.005213	0.000385
Story2	5.9182	Top	0.003716	0.000214
Story1	3.048	Top	0.000125	0.000017
Base	0	Top	0	0

**The maximum Drift =
0.00521 < 0.006
“SAFE”**

Story Response - Maximum Story Drifts

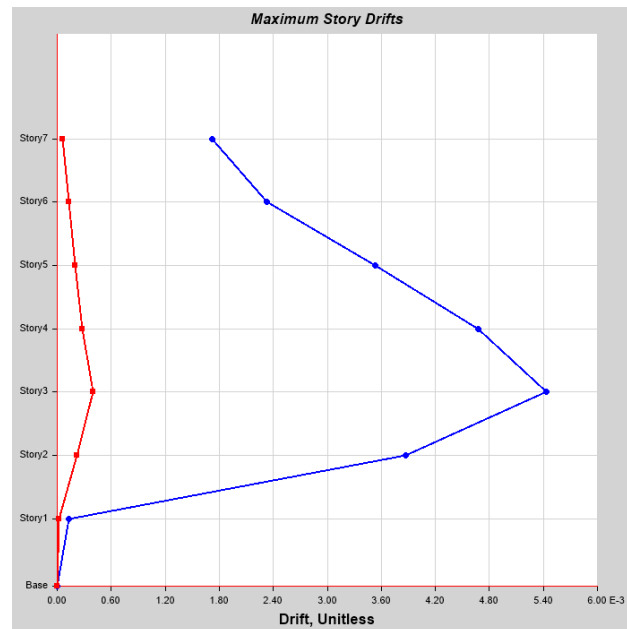
Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp1		
Display Type	Max story drifts	Story Range	All Stories
Load Case	EQx ULS	Top Story	Story7
Output Type	MaxMin	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir	Y-Dir
	m			
Story7	20.2692	Top	0.001722	0.000064
Story6	17.399	Top	0.002328	0.000125
Story5	14.5288	Top	0.003536	0.000203
Story4	11.6586	Top	0.004677	0.00028
Story3	8.7884	Top	0.00543	0.000401
Story2	5.9182	Top	0.003871	0.000223
Story1	3.048	Top	0.000131	0.000018
Base	0	Top	0	0

The maximum Drift ratio = 0.00543,

Building to be safe under ULS : ETABS Result * Ductility Factor * Scale Factor < 0.025

$$0.00543 * 4 * 0.88 = 0.01911 < 0.025, \text{ "SAFE"}$$

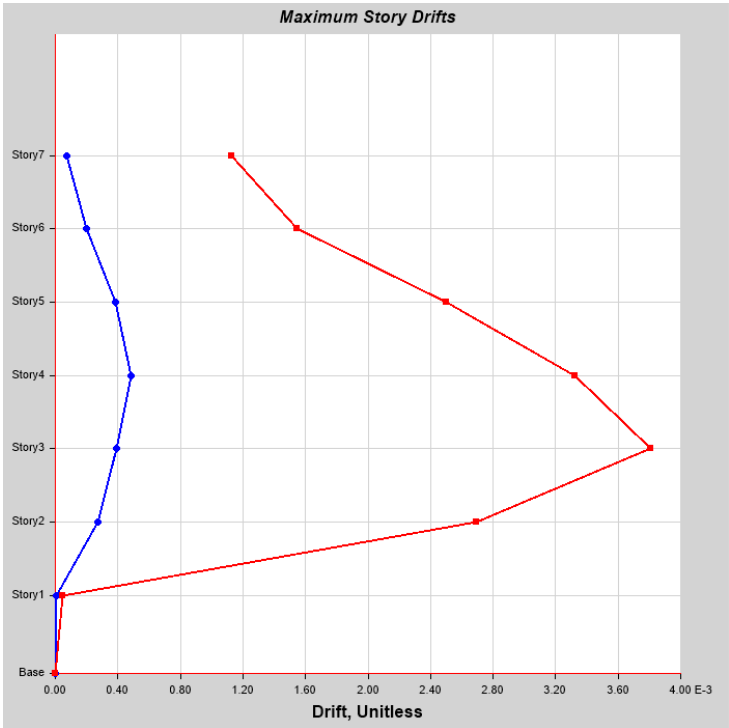
Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp5		
Display Type	Max story drifts	Story Range	All Stories
Load Case	EQy SLS	Top Story	Story7
Output Type	Step Number 1	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir	Y-Dir
	m			
Story7	20.2692	Top	0.00007	0.00113
Story6	17.399	Top	0.000202	0.001543
Story5	14.5288	Top	0.000388	0.002501
Story4	11.6586	Top	0.000484	0.003325
Story3	8.7884	Top	0.000394	0.003806
Story2	5.9182	Top	0.000275	0.002692
Story1	3.048	Top	0.000006	0.000045
Base	0	Top	0	0

The maximum Drift Ratio = 0.00398< 0.006
“SAFE”

Story Response - Maximum Story Drifts

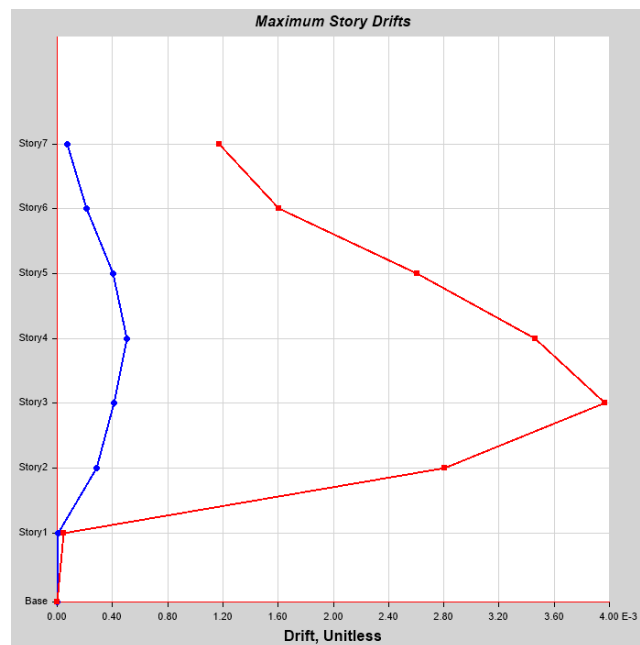
Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp6		
Display Type	Max story drifts	Story Range	All Stories
Load Case	EQy ULS	Top Story	Story7
Output Type	Step Number 1	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir	Y-Dir
	m			
Story7	20.2692	Top	0.000073	0.001177
Story6	17.399	Top	0.000211	0.001607
Story5	14.5288	Top	0.000404	0.002605
Story4	11.6586	Top	0.000504	0.003463
Story3	8.7884	Top	0.000411	0.003964
Story2	5.9182	Top	0.000287	0.002804
Story1	3.048	Top	0.000006	0.000047
Base	0	Top	0	0

The maximum Drift =
 $0.00364 < 0.006$
SAFE

Building to be safe under ULS : ETABS Result * Ductility Factor * Scale Factor < 0.025
 $0.00364 * 4 * 0.88 = 0.01281 < 0.025$, “SAFE”

Storey displacement is the absolute lateral movement of a floor level with respect to the base of the structure under seismic or lateral loading. It represents how far a particular storey shifts from its original position during analysis. As per **NBC 105: 2025**, storey displacement is evaluated to understand the overall sway behavior and stability of the building under earthquake effects. Controlling storey displacement is important to ensure structural safety, maintain serviceability, and prevent excessive overall movement that may lead to instability or functional problems. ETABS analysis directly provides storey displacement values for each level, which can be used to assess global building response and verify compliance with design requirements.

The Structure to be safe under Displacement Check :

Under SLS:

$$\frac{\text{Maximum Displacement}}{\text{Storey Height}} \leq 0.006$$

$$\text{Maximum Displacement} \leq 0.006 * \text{Storey Height}$$

$$\text{Maximum Displacement} \leq 0.006 * (16.9925 * 1000)$$

$$\text{Maximum Displacement (mm)} \leq 101.955 \text{ mm}$$

Under ULS:

$$\text{Maximum Displacement (mm)} * \text{Ductility Factor} * \text{Scale factor}$$

$$\leq 0.025 * \text{Storey Height}$$

$$\leq 0.025 * (16.9925 * 1000)$$

$$\leq 424.812 \text{ mm}$$

$$\text{Maximum Displacement (mm)} * \text{Ductility Factor} * \text{Scale factor} \leq 424.81 \text{ mm}$$

Story Response - Maximum Story Displacement

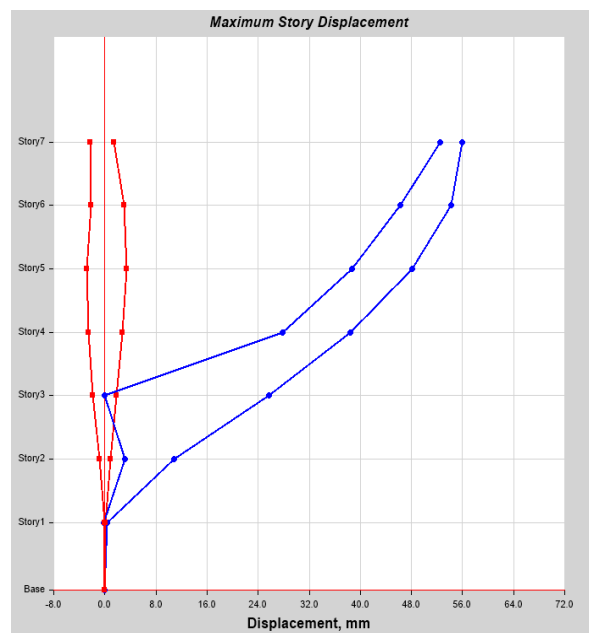
Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp2		
Display Type	Max story displ	Story Range	All Stories
Load Case	EQx SLS	Top Story	Story7
Output Type	MaxMin	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir Max	X-Dir Min	Y-Dir Max	Y-Dir Min
	m		mm	mm	mm	mm
Story7	20.2692	Top	55.977	52.491	1.368	-2.341
Story6	17.399	Top	54.224	46.289	3.049	-2.181
Story5	14.5288	Top	48.061	38.687	3.388	-2.762
Story4	11.6586	Top	38.433	27.763	2.735	-2.514
Story3	8.7884	Top	25.664	0	1.799	-1.941
Story2	5.9182	Top	10.894	3.08	0.858	-0.804
Story1	3.048	Top	0.387	-0.188	0.106	-0.037
Base	0	Top	0	0	0	0

Max Displacement = 55.976 mm < 101.955 mm “SAFE”

Story Response - Maximum Story Displacement

Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

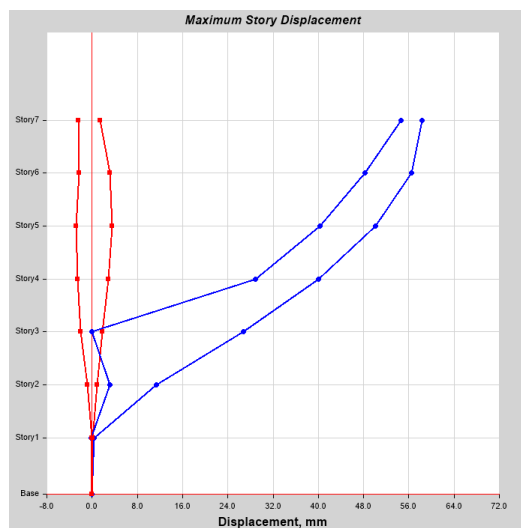
Name StoryResp2

Display Type Max story displ Story Range All Stories

Load Case EQx ULS Top Story Story7

Output Type MaxMin Bottom Story Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir Max	X-Dir Min	Y-Dir Max	Y-Dir Min
	m		mm	mm	mm	mm
Story7	20.2692	Top	58.309	54.678	1.425	-2.439
Story6	17.399	Top	56.483	48.218	3.176	-2.272
Story5	14.5288	Top	50.064	40.299	3.529	-2.877
Story4	11.6586	Top	40.034	28.919	2.849	-2.619
Story3	8.7884	Top	26.734	0	1.874	-2.022
Story2	5.9182	Top	11.348	3.208	0.894	-0.838
Story1	3.048	Top	0.403	-0.196	0.111	-0.039
Base	0	Top	0	0	0	0

Max Storey Displacement = 58.309 mm

Maximum Displacement (mm) * Ductility Factor * Scale factor =< 424.81 mm

58.309 * 4 * 0.88 = 205.24 mm < 424.81 mm

Story Response - Maximum Story Displacement

Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp2		
Display Type	Max story displ	Story Range	All Stories
Load Case	EQy SLS	Top Story	Story7
Output Type	MaxMin	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir Max	X-Dir Min	Y-Dir Max	Y-Dir Min
	m		mm	mm	mm	mm
Story7	20.2692	Top	0.772	-4.003	40.235	35.28
Story6	17.399	Top	7.34	-6.791	41.509	30.59
Story5	14.5288	Top	6.495	-6.015	37.849	26.084
Story4	11.6586	Top	4.927	-4.674	30.115	19.404
Story3	8.7884	Top	2.912	-2.909	19.859	0
Story2	5.9182	Top	1.362	-1.162	8.303	2.433
Story1	3.048	Top	0.028	-0.027	0.146	0.06
Base	0	Top	0	0	0	0

Max Displacement = 41.509 mm < 101.955 mm “ SAFE”

Story Response - Maximum Story Displacement

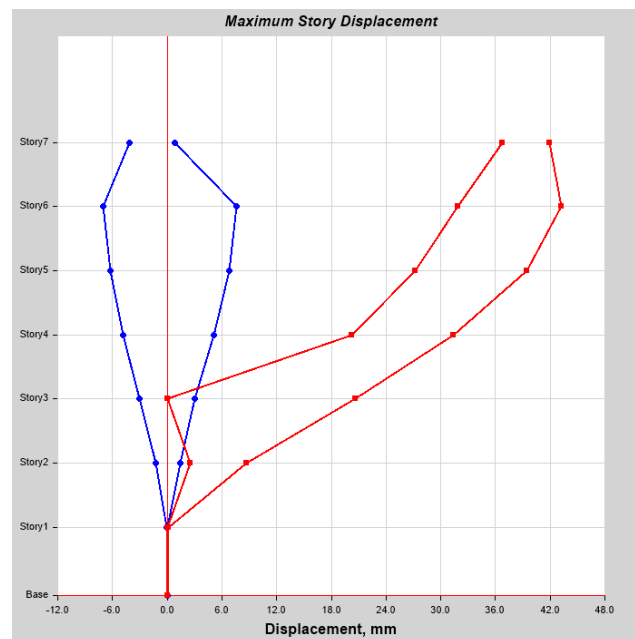
Summary Description

This is story response output for a specified range of stories and a selected load case or load combination.

Input Data

Name	StoryResp2		
Display Type	Max story displ	Story Range	All Stories
Load Case	EQy ULS	Top Story	Story7
Output Type	MaxMin	Bottom Story	Base

Plot



Tabulated Plot Coordinates

Story Response Values

Story	Elevation	Location	X-Dir Max	X-Dir Min	Y-Dir Max	Y-Dir Min
	m		mm	mm	mm	mm
Story7	20.2692	Top	0.804	-4.17	41.911	36.75
Story6	17.399	Top	7.646	-7.074	43.239	31.865
Story5	14.5288	Top	6.765	-6.266	39.426	27.171
Story4	11.6586	Top	5.132	-4.869	31.369	20.213
Story3	8.7884	Top	3.033	-3.03	20.686	0
Story2	5.9182	Top	1.418	-1.21	8.649	2.535
Story1	3.048	Top	0.029	-0.028	0.152	0.062
Base	0	Top	0	0	0	0

Max Storey Displacement = 43.238 mm

Maximum Displacement (mm) * Ductility Factor * Scale factor =< 424.81 mm

43.238 * 4 * 0.88 = 152.19 mm < 424.81 mm

The **centre of mass (CM) and centre of rigidity (CR)** were evaluated for all storeys in both principal directions. The eccentricity in each direction was computed as the ratio of the distance between CM and CR to the corresponding plan dimension of the building.

The calculated eccentricity ratios e_x and e_y for all storeys were found to be less than 5% of the respective plan dimensions.

According to the provisions of **NBC 105:2025**, a building is classified as torsionally irregular when the eccentricity exceeds 5% of the plan dimension in the direction under consideration. Since the computed eccentricities are within the permissible limit, the building is classified as **torsionally regular in plan**.

Therefore, the structure does not exhibit significant torsional irregularity, and no additional manual eccentricity was required to be assigned in the analytical model. The default accidental eccentricity considered in the software was deemed adequate for seismic analysis.

Torsional irregularity check

Torsional irregularity of the structure was evaluated in accordance with the provisions of **NBC 105:2025 – Seismic Design of Buildings in Nepal**. As per the code, torsional irregularity at a particular storey is considered to exist when the ratio of the maximum horizontal displacement at one extreme end of the storey to the minimum horizontal displacement at the opposite end exceeds **1.5** under seismic loading.

The torsional irregularity ratio is calculated as:

$$\text{Torsional Ratio} = \frac{\Delta_{max}}{\Delta_{min}}$$

Where:

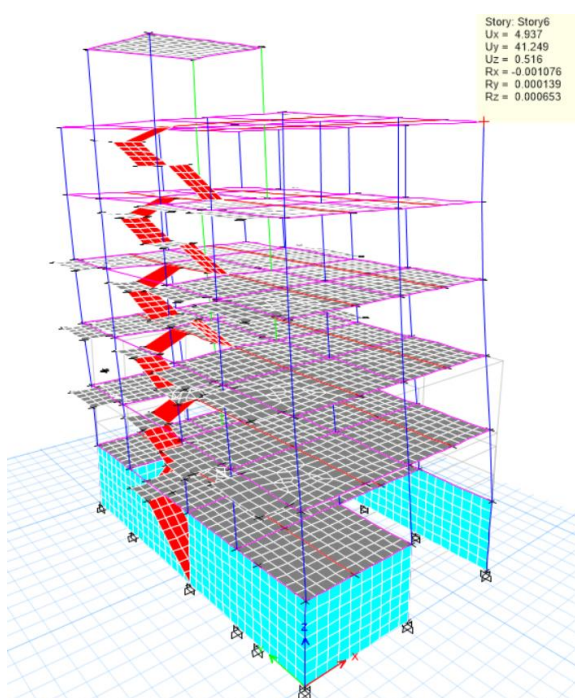
- Δ_{max} = Maximum lateral displacement at extreme end of storey
- Δ_{min} = Minimum lateral displacement at opposite extreme end

As per NBC 105:2025, torsional irregularity is present if:

$$\frac{\Delta_{max}}{\Delta_{min}} > 1.5$$

Since the calculated torsional ratios in both X and Y directions are less than 1.5, the structure does **not exhibit torsional irregularity** at the terrace level.

Therefore, the building is considered torsionally regular at this storey. Similar checks were performed for other storeys, and compliance with code requirements was ensured.



Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	1	1
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	0.267	0.133
Maximum Displacement (Δ_{max})	0.333	0.146
$(\Delta_{max})/(\Delta_{min})$	1.247	1.097
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	2	2
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	9.865	6.6911
Maximum Displacement (Δ_{max})	10.623	8.185
$(\Delta_{max})/(\Delta_{min})$	1.02	1.222
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	3	3
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	23.443	16.614
Maximum Displacement (Δ_{max})	25.063	19.572
$(\Delta_{max})/(\Delta_{min})$	1.069	1.178
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	4	4
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	35.603	24.518
Maximum Displacement (Δ_{max})	37.501	29.661
$(\Delta_{max})/(\Delta_{min})$	1.053	1.209
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	5	5
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	45.435	30.41
Maximum Displacement (Δ_{max})	46.892	37.281
$(\Delta_{max})/(\Delta_{min})$	1.032	1.22
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	6(Terrace)	6 (Terrace)
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	51.405	34.932
Maximum Displacement (Δ_{max})	52.931	41.249
$(\Delta_{max})/(\Delta_{min})$	1.029	1.18
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Check For Torsional Irregularity		
	Along X-axis	Along Y-axis
Storey Level	7	7
Load Case	EQx ULS	EQy ULS
Minimum Displacement (Δ_{min})	56.494	38.232
Maximum Displacement (Δ_{max})	57.162	41.548
$(\Delta_{max})/(\Delta_{min})$	1.011	1.086
Allowable $(\Delta_{max})/(\Delta_{min})$	1.5	1.5
	OK	OK

Integrated wall reaction and Base Moment

TABLE: Integrated Wall Reactions								
Story	WallBay	UniqueName	Output Case	Case Type	Step Type	FX	FY	FZ
						kN	kN	kN
Story1	W4	39	Envelope	Combination	Max	88.9621	267.568	205.7765
Story1	W13	90	Envelope	Combination	Max	797.7391	56.918	1654.217
Story1	W14	91	Envelope	Combination	Max	845.9647	58.7476	1686.2794
Story1	W15	92	Envelope	Combination	Max	20.0733	181.2635	1708.9522
Story1	W16	93	Envelope	Combination	Max	56.5388	375.0564	2141.8199
Story1	W17	99	Envelope	Combination	Max	25.4505	379.3847	1478.0236
Story1	W18	100	Envelope	Combination	Max	10.0889	352.4046	1472.0737
Story1	W19	5	Envelope	Combination	Max	181.2241	169.5975	1574.2656
Story1	W20	37	Envelope	Combination	Max	137.6077	268.0283	1375.8333
Story1	W21	60	Envelope	Combination	Max	194.0647	543.161	1781.4351
Story1	W22	89	Envelope	Combination	Max	85.5902	134.3269	1548.6479
Story1	W23	101	Envelope	Combination	Max	1232.9971	26.3013	1997.6389
Story1	W10	141	Envelope	Combination	Max	8.5463	105.3003	377.8641

BASE MOMENTS:

Story	WallBay	UniqueName	Output Case	Case Type	Step Type	MX	MY	MZ
						kN-m	kN-m	kN-m
Story 1	W4	39	Envelope	Combination	Max	20.1324	57.9859	-12.0537
Story 1	W13	90	Envelope	Combination	Max	20.5683	822.8703	34.5313
Story 1	W14	91	Envelope	Combination	Max	19.5868	612.4755	75.6699
Story 1	W15	92	Envelope	Combination	Max	73.947	16.0226	49.0346
Story 1	W16	93	Envelope	Combination	Max	373.1139	47.257	2.5263
Story 1	W17	99	Envelope	Combination	Max	179.3603	21.8822	6.1165
Story 1	W18	100	Envelope	Combination	Max	316.1547	8.3102	74.9679
Story 1	W19	5	Envelope	Combination	Max	315.4702	108.6893	22.7234

Story 1	W20	37	Envelope	Combination	Max	150.4534	76.1835	7.887
Story 1	W21	60	Envelope	Combination	Max	233.7651	103.4533	15.7298
Story 1	W22	89	Envelope	Combination	Max	198.6035	43.6286	66.8038
Story 1	W23	101	Envelope	Combination	Max	28.4499	1286.4368	58.5371
Story 1	W10	141	Envelope	Combination	Max	29.3406	5.5019	3.8752

Modal Period and Frequencies

TABLE: Modal Periods And Frequencies					
Case	Mode	Period	Frequency	CircFreq	Eigenvalue
		sec	cyc/sec	rad/sec	rad ² /sec ²
Modal	1	1.011	0.989	6.214	38.6141
Modal	2	0.896	1.116	7.009	49.1259
Modal	3	0.806	1.24	7.7936	60.7401
Modal	4	0.317	3.153	19.812	392.5141
Modal	5	0.282	3.552	22.3188	498.1295
Modal	6	0.259	3.867	24.3	590.4895
Modal	7	0.183	5.474	34.3964	1183.1148
Modal	8	0.168	5.954	37.4082	1399.3735
Modal	9	0.15	6.68	41.9706	1761.5292
Modal	10	0.134	7.455	46.8391	2193.9053
Modal	11	0.125	8.014	50.3547	2535.5921
Modal	12	0.117	8.547	53.703	2884.0147
Modal	13	0.1	10.05	63.1468	3987.5241
Modal	14	0.096	10.441	65.6039	4303.868
Modal	15	0.089	11.245	70.653	4991.8459
Modal	16	0.077	12.999	81.6766	6671.075
Modal	17	0.075	13.342	83.8281	7027.1458
Modal	18	0.07	14.374	90.3124	8156.3273
Modal	19	0.036	28	175.9292	30951.1005
Modal	20	0.025	39.344	247.206	61110.797

TABLE: Modal Participating Mass Ratios										Modal Participating Mass Ratios				RZ		Sum
Case	Mode	Period sec	UX	UY	UZ	SumUX	SumUY	SumUZ						RX	RY	
Modal	1	1.011	0.6738	0.00004457	0	0.6738	0.00004457	0	0.00003611					0.00003611	0.4167	0.0000
Modal	2	0.896	0.0009	0.4288	0	0.6747	0.4288	0	0.2499					0.2499	0.0013	0.2433
Modal	3	0.806	0.0007	0.2485	0	0.6754	0.6773	0	0.1662					0.1662	0.0015	0.4159
Modal	4	0.317	0.0839	0.00001727	0	0.7594	0.6773	0	0.00003971					0.00003971	0.2312	0.00004224
Modal	5	0.282	0.0001	0.0631	0	0.7594	0.7404	0	0.1759					0.1759	0.00001638	0.022
Modal	6	0.259	0.0001	0.0152	0	0.7595	0.7556	0	0.0571					0.0571	0.0002	0.0615
Modal	7	0.183	0.0285	0.0001	0	0.788	0.7558	0	0.0001					0.0001	0.0182	0.0004
Modal	8	0.168	0.0002	0.0237	0	0.7882	0.7795	0	0.0148					0.0148	0.0001	0.0009
Modal	9	0.15	0.0012	0.0028	0	0.7894	0.7823	0	0.0022					0.0022	0.0013	0.0272
Modal	10	0.134	0.0171	0.001	0	0.8065	0.7833	0	0.0008					0.0008	0.02	0.0003
Modal	11	0.125	0.0001	0.0164	0	0.8067	0.7997	0	0.0166					0.0166	0.0002	0.0026
Modal	12	0.117	0.0001	0.0001	0	0.8067	0.7998	0	0.0001					0.0001	0.0001	0.0081
Modal	13	0.1	0.0145	0.0001	0	0.8212	0.7999	0	0.0002					0.0002	0.0206	0.0007
Modal	14	0.096	0.0004	0.0101	0	0.8216	0.81	0	0.0146					0.0146	0.0005	0.003
Modal	15	0.089	0.0001	0.0029	0	0.8217	0.813	0	0.0039					0.0039	0.0001	0.0106
Modal	16	0.077	0.0045	0.000006589	0	0.8261	0.813	0	0.00001218					0.00001218	0.0051	0.00004665
Modal	17	0.075	0.00000197	0.0028	0	0.8261	0.8158	0	0.0027					0.0027	0	0.0006
Modal	18	0.07	0.00000165	0.0003	0	0.8261	0.816	0	0.0002					0.0002	0.000001817	0.0035
Modal	19	0.036	0.1631	0.0002	0	0.9892	0.8162	0	0.0003					0.0003	0.2656	0.0041
Modal	20	0.025	0.005	0.00002159	0	0.9942	0.8163	0	0.00003449					0.00003449	0.0082	0.0086

RZ	SumRX	SumRY	SumRZ
0.0031	0.00003611	0.4167	0.0031
0.2433	0.2499	0.418	0.2464
0.4159	0.4161	0.4195	0.6623
0.00004224	0.4161	0.6507	0.6623
0.022	0.592	0.6507	0.6843
0.0615	0.649	0.6509	0.7459
0.0004	0.6492	0.6691	0.7462
0.0009	0.6639	0.6693	0.7471
0.0272	0.6661	0.6705	0.7744
0.0003	0.6669	0.6905	0.7747
0.0026	0.6835	0.6907	0.7773
0.0081	0.6835	0.6908	0.7854
0.0007	0.6837	0.7114	0.7862
0.003	0.6984	0.7119	0.7891
0.0106	0.7023	0.712	0.7997
0.00004665	0.7023	0.7171	0.7998
0.0006	0.705	0.7171	0.8004
0.0035	0.7053	0.7171	0.804
0.0041	0.7056	0.9827	0.8081
0.0086	0.7056	0.9909	0.8167

Load Pattern Definitions - Auto Seismic - User Coefficient

Name	Ecc Ratio	Top Story	Bottom Story	C	K	Weight Used	Base Shear
						kN	kN
EQx SLS	0.05	Story7	Story1	0.126	1.067		
EQx SLS(1/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQx SLS(2/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQx SLS(3/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQx ULS	0.05	Story7	Story1	0.13125	1.067		
EQx ULS(1/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217
EQx ULS(2/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217
EQx ULS(3/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217
EQy SLS	0.05	Story7	Story1	0.126	1.067		
EQy SLS(1/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQy SLS(2/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQy SLS(3/3)	0.05	Story7	Story1	0.126	1.067	17015.0224	2143.8928
EQy ULS	0.05	Story7	Story1	0.13125	1.067		
EQy ULS(1/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217
EQy ULS(2/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217
EQy ULS(3/3)	0.05	Story7	Story1	0.13125	1.067	17015.0224	2233.2217

CHAPTER 5 : DETAIL DESIGN OF MAIN STRUCTURAL COMPONENTS

The design process must consider durability, construction practices, and service conditions as an integrated whole. Achieving the intended design objectives requires strict compliance with established standards related to materials, workmanship, maintenance, and the proper use of the structure during its service life.

This chapter presents the complete design procedure along with sample calculations for individual structural elements such as slabs, beams, columns, staircases, basement walls, lift walls, ribbed slabs, and mat foundations.

The elements covered are:

- Slab Design
- Beam Design
- Column Design
- Staircase Design
- Basement Wall Design
- Mat Foundation Design

5.1 Design of slab

The slab is designed as a **monolithic element**. Clause **34.2 of IS 456:2000** has been followed for slab design. Bending moments in slabs constructed monolithically with supports have been calculated by assuming the slab to act as a continuous member over supports or capable of free rotation, similar to members in a continuous framework, while accounting for the stiffness of the supports.

A slab is a rigid plate that deflects when subjected to floor loads during the construction stage. In such conditions, all points on the slab are displaced equally when the load is applied. Structurally, a slab behaves as a flexural element and is mainly classified into **one-way slabs** and **two-way slabs**, based on the ratio of the longer span to the shorter span.

One-way slabs are those in which the longer span is more than **twice** the shorter span. A continuous one-way slab is analyzed in a manner similar to a continuous beam.

Two-way slabs are slabs supported on **all four sides**, allowing load distribution in two directions. The present project involves the design of two-way slabs, and the general analysis has been carried out in accordance with **Clause 26.2.3 of IS 456:2000**. Compared to one-way slabs, deflections and bending moments in two-way slabs are significantly

reduced.

In this project, the slabs have been designed considering **full edge restraint**. The degree of restraint may vary depending on whether the slab is continuous over its supports or cast monolithically with the supporting beams. Due to the presence of restraint, **hogging or negative bending moments** develop near the top face of the slab at the supports. In such slabs, **corner lifting is prevented**, and therefore no torsional reinforcement is required at the corners.

Calculation of slab :

Reference	Steps	Calculation	Output
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Slab ID: (1-2, E-D) . Two adjacent edges discontinuous IS 456;2000 Table 26 IS 456;2000 ANNEX D-1.1	1. 2. 3.	Design Constants : Effective spans: L_x = 3210 mm L_y = 4903 mm Provided effective depth, d = 130 mm Effective cover = 15 mm Overall depth, D = 150 mm Span ratio check: l_y / l_x = 1.531 < 2 ∴ Two-way slab Self-weight of slab = 3.75 kN/m² Live load = 2.0 kN/m² Floor finish = 1 kN/m² Total load = 6.75 kN/m² Considering 1000 mm wide strip along shorter span: Design load: W_u = 1.5 (DL + LL) = 1.5 × 6.75 = 10.125kN/m Moment Calculation : Shorter span coefficient α_x⁻ = 0.0757 Shorter span coefficient α_x⁺ = 0.0566 Longer span coefficient α_y⁻ = 0.0470 Longer span coefficient α_y⁺ = 0.0350 For Shorter Span: Support moment (m_x⁻) = α_x⁻ W_u leffx² = 9.016 KNm Mid Span moment (m_x⁺) = α_x⁺ W_u leffx² = 6.7418 KNm For Longer Span: Support moment (m_y⁻) = α_y⁻ W_u leffx² = 5.5983 KNm Mid Span moment (m_y⁺) = α_y⁺ W_u leffx² = 4.196 KNm	
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Reference	Steps	Calculation	Output
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Slab ID: (1-2, E-D) . Two adjacent edges discontinuous	7.	Along Longer span: a) Reinforcement in the middle strip $M_{uy}^+ = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 75.0067 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1ϕ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280$ mm Adopted Spacing = 250 mm $A_{ty}^+ \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$ b) Reinforcement in the Support strip $M_{uy}^- = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 100.552 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1ϕ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280$ mm Adopted Spacing = 250 mm $A_{ty}^- \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$ Check for shear $\begin{aligned} \text{Factored shear force (V}_u) &= \frac{w_x \cdot l_x}{2} \\ &= \frac{w \cdot l_x \cdot e f f^2}{2} * \frac{r^4}{1+r^4} \\ &= 13.748 \text{ KN} \end{aligned}$ $\text{Nominal shear stress } (\tau_v) = \frac{V_u}{b d} = 0.106 \text{ N/mm}^2$	Provide 8mm dia bar with Spacing 250mm Provide 8mm dia bar with Spacing 250mm
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Reference	Steps	Calculation	Output
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Slab ID: (1-2, E-D) . Two adjacent edges discontinuous IS 456 : 2000 Table 20	8.	$\%A_t = \frac{Atx \cdot 100}{bd} = 0.154$ $\tau_c \text{ max} = 3.1 \text{ N/mm}^2$ $\tau_c \text{ for M25} = 0.243 \text{ N/mm}^2$ Here , $\tau_v < \tau_c$, $\tau_v < 1/2 \tau_c \text{ max}$ Hence safe in shear . Check in Deflection $(\frac{L}{d})_{\text{allowable}} = \alpha \beta \gamma \delta \lambda = 46.8$ $(\frac{L}{d})_{\text{provided}} = 24.62 < (\frac{L}{d})_{\text{allowable}}$	OK
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Calculation of slab :

Reference	Steps	Calculation	Output
Slab ID: (2-4, A-B) . Two adjacent edges discontinuous	1.	Design Constants : Effective spans: Lx = 3607 mm Ly = 4776 mm Provided effective depth, d = 130 mm Effective cover = 15 mm Overall depth, D = 150 mm Span ratio check: ly / lx = 1.324 < 2 ∴ Two-way slab	
	2.	Self-weight of slab = 3.75 kN/m ² Live load = 2.0 kN/m ² Floor finish = 1 kN/m ² Total load = 6.75 kN/m ² Considering 1000 mm wide strip along shorter span: Design load: Wu = 1.5 (DL + LL) = 1.5 × 6.75 = 10.125kN/m	
IS 456;2000 Table 26	3.	Moment Calculation : Shorter span coefficient $\alpha x^- = 0.066$ Shorter span coefficient $\alpha x^+ = 0.047$ Longer span coefficient $\alpha y^- = 0.0500$ Longer span coefficient $\alpha y^+ = 0.0350$ For Shorter Span: Support moment ($m x^-$) = $\alpha x^- W_u \text{ leff} x^2$ = 8.694 KNm Mid Span moment ($m x^+$) = $\alpha x^+ W_u \text{ leff} x^2$ = 6.586 KNm For Longer Span: Support moment ($m y^-$) = $\alpha y^- W_u \text{ leff} x^2$ = 6.191 KNm Mid Span moment ($m y^+$) = $\alpha y^+ W_u \text{ leff} x^2$ = 4.661 KNm	
IS 456;2000 ANNEX D-1.1			

[illegible]

Reference	Steps	Calculation	Output
Slab ID: (2-4, A-B) . Two adjacent edges discontinuous		Along Longer span: a) Reinforcement in the middle strip $M_{uy}^+ = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 82.58 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1φ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280$ mm Adopted Spacing = 250 mm $A_{ty}^+ \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$ b) Reinforcement in the Support strip $M_{uy}^- = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 111.387 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1φ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280$ mm Adopted Spacing = 250 mm $A_{ty}^- \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$	Provide 8mm dia bar with Spacing 250mm Provide 8mm dia bar with Spacing 250mm
IS 456: 2000 CL 40.2.3	7.	Check for shear $\text{Factored shear force (V}_u\text{)} = \frac{w_x \cdot l_x}{2}$ $= \frac{w \cdot l_{xeff}}{2} \cdot \frac{r^4}{1+r^4}$ $= 13.77 \text{ KN}$ $\text{Nominal shear stress } (\tau_v) = \frac{V_u}{b d} = 0.105 \text{ N/mm}^2$	

Reference	Steps	Calculation	Output
Slab ID: (2-4, A-B) . Two adjacent edges discontinuous IS 456 : 2000 Table 20 IS 456:2000 CL 23.2.1	8.	$\%A_t = \frac{Atx - *100}{bd} = 0.154$ $\tau_c \text{ max} = 3.1 \text{ N/mm}^2$ $\tau_c \text{ for M25} = 0.243 \text{ N/mm}^2$ Here , $\tau_v < \tau_c$, $\tau_v < 1/2 \tau_c \text{ max}$ Hence safe in shear . Check in Deflection $\left(\frac{L}{d}\right)_{\text{allowable}} = \alpha \beta \gamma \delta \lambda = 46.8$ $\left(\frac{L}{d}\right)_{\text{provided}} = 27.74 < \left(\frac{L}{d}\right)_{\text{allowable}}$	OK

Calculation of slab :

Reference	Steps	Calculation	Output
Slab ID: (2-4, D-C) . One short edge discontinuous	1.	Design Constants : Effective spans: $L_x = 4725 \text{ mm}$ $L_y = 4776 \text{ mm}$ Provided effective depth, $d = 130 \text{ mm}$ Effective cover = 15 mm Overall depth, $D = 150 \text{ mm}$ Span ratio check: $l_y / l_x = 1.01 < 2$ $\therefore$ Two-way slab	
IS 456;2000 Table 26	2.	Self-weight of slab = 3.75 kN/m^2 Live load = 2.0 kN/m^2 Floor finish = 1 kN/m^2 Total load = 6.75 kN/m^2 Considering 1000 mm wide strip along shorter span: Design load: $W_u = 1.5 (\text{DL} + \text{LL})$ $= 1.5 \times 6.75 = 10.125 \text{ kN/m}$	
IS 456;2000 ANNEX D-1.1	3.	Moment Calculation : Shorter span coefficient $\alpha_x^- = 0.0376$ Shorter span coefficient $\alpha_x^+ = 0.0284$ Longer span coefficient $\alpha_y^- = 0.037$ Longer span coefficient $\alpha_y^+ = 0.028$ For Shorter Span: Support moment (m_x^-) = $\alpha_x^- W_u l_{\text{eff}}^2$ $= 8.499 \text{ KNm}$ Mid Span moment (m_x^+) = $\alpha_x^+ W_u l_{\text{eff}}^2$ $= 6.419 \text{ KNm}$ For Longer Span: Support moment (m_y^-) = $\alpha_y^- W_u l_{\text{eff}}^2$ $= 6.329 \text{ KNm}$ Mid Span moment (m_y^+) = $\alpha_y^+ W_u l_{\text{eff}}^2$ $= 8.363 \text{ KNm}$	

[illegible]

Reference	Steps	Calculation	Output
Slab ID: (2-4, D-C) . Two adjacent edges discontinuous		Along Longer span: a) Reinforcement in the middle strip $M_{uy}^+ = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 113.91 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1φ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280 \text{ mm}$ Adopted Spacing = 250 mm $A_{ty}^+ \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$ b) Reinforcement in the Support strip $M_{uy}^- = 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \frac{A_{st} \cdot f_y}{b d f_{ck}}\right)$ Ast required = 151.41 mm², Adpot min area = 180mm² Provide 8mm dia bar , A1φ = 50.26 mm² Spacing = $\frac{50.26 \cdot 1000}{180} = 280 \text{ mm}$ Adopted Spacing = 250 mm $A_{ty}^- \text{ Provided} = \frac{50.26 \cdot 1000}{250} = 201.04 \text{ mm}^2$	Provide 8mm dia bar with Spacing 250mm Provide 8mm dia bar with Spacing 250mm
IS 456: 2000 CL 40.2.3	7.	Check for shear $\text{Factored shear force (V}_u) = \frac{w_x \cdot l_x}{2}$ $= \frac{w \cdot l_{xeff}}{2} \cdot \frac{r^4}{1+r^4}$ $= 12.19 \text{ KN}$ $\text{Nominal shear stress } (\tau_v) = \frac{V_u}{b d} = 0.0937 \text{ N/mm}^2$	

Reference	Steps	Calculation	Output
Slab ID: (2-4, D-C) . Two adjacent edges discontinuous IS 456 : 2000 Table 20 IS 456:2000 CL 23.2.1	8.	$\%A_t = \frac{A_{tx} \times 100}{bd} = 0.154$ $\tau_c \text{ max} = 3.1 \text{ N/mm}^2$ $\tau_c \text{ for M25} = 0.243 \text{ N/mm}^2$ Here , $\tau_v < \tau_c$, $\tau_v < 1/2 \tau_c \text{ max}$ Hence safe in shear . Check in Deflection $\left(\frac{L}{d}\right)_{\text{allowable}} = \alpha \beta \gamma \delta \lambda = 46.8$ $\left(\frac{L}{d}\right)_{\text{provided}} = 36.34 < \left(\frac{L}{d}\right)_{\text{allowable}}$	OK

5.2 Design Of Column

Columns are the vertical members that transfer axial as well as lateral loads and moments from the upper structure. Columns support the beams and slabs, and transfer the forces from beams to the foundation.

Depending upon the slenderness ratio, column is divided into short column (*slenderness ratio* ≤ 12) and long column (*slenderness ratio* > 12) which fails predominantly by compression and buckling respectively. The slenderness ratio of a column is given by the ratio of effective length to dimension of the column.

Based on loading criteria, the columns can be axially loaded, uniaxially loaded or biaxially loaded columns, which is classified according to the eccentricity ratio of the column in two orthogonal directions.

The following steps are followed while designing the column:

Step 1: Preliminary Design

The preliminary design of the column is done in order to fix the size of columns for the preliminary analysis of the structure. Preliminary design of column is done considering the factored axial compressive load due to dead and imposed load only.

$$\text{Factored Load, } P_u = 1.5 \times P$$

Ultimate load carrying capacity,

$$P_u = 0.4 f_{ck} A_c + 0.67 f_y A_s$$

$$P_u = 0.4 f_{ck} A \left(1 - \frac{P}{P_u}\right) + 0.67 f_y A_p \left(\frac{P}{P_u}\right)$$

Where , P/P_u = Working Stress and Factored ultimate load carrying capacity

f_{ck} = characteristic strength of concrete

f_y = yield strength of longitudinal reinforcement

A_c, A_s, A = Area of concrete, steel and gross area of column

p = percentage of longitudinal rebars

Step 2 : Slenderness Check

Slenderness ratio is checked based on the sway or no-sway condition according to fig. 27 of IS 456:2000 and clause number 25.1.2 of IS 456:2000. Slenderness ratio is needed to determine the mode of failure and additional moment due to $P - \Delta$ effect.

Step 3: Check for eccentricity

Eccentricity check is required in order to decide the condition of loading in the column, axial, uniaxial or biaxially loaded column. When the eccentricity exceeds over $0.05D$, the column has significant moment as well and design should be conducted according to the interaction diagram of SP 16.

Step 4: Design of Longitudinal Reinforcement

The longitudinal reinforcement is fixed based by calculation in uniaxially loaded column while iteration should be done for biaxially loaded column, each time assuming certain percentage of reinforcement.

Step 5: Design of Lateral Reinforcement

The design of lateral tie is required to provide appropriate ductile requirement as well as safety against shear failure. Special lateral ties are provided wherever necessary as given by the code IS 13920.

The sample calculation for manual design of a typical column is shown below in table:

Design Steps	References	Calculations	Remark
1.	NBC 105: 2025	<u>Member Dimensions</u> Width = 500 mm > 300mm Depth = 500 mm > 300mm Check : 20*16 = 320mm < 500 mm B/D = 1 > 0.45 Assume, Clear Cover = 50 mm Preliminary diameter of L Bar = 16mm Effective Cover = $40 + 16/2 + 8 =$ 56 mm	B,D> 300mm OK 20*do < 500mm OK
2.		<u>Material Properties</u> Grade of Concrete (fck) = M25 Grade of Steel (fy) = Fe 500	
3.		<u>Member Properties</u> Length of Column = 2.870 m Depth of beam = 350mm Effective Length Factor : Kx = 0.652 (M3) Ky = 0.639 (M2) Unsupported Length = 2.870-0.350 = 2.520 mm Effective Length of Column (Lex) = Unsupp length * Kx = 2.520*0.934	

4.		$= 2.353 \text{ mm}$ Similarly , $L_{ey} = 2.142 \text{ mm}$ <u>Load Calculation</u> Axial load on Column = 2111.46 KN We take Design Load as 2111.46 KN Moment about x-axis $M_{ux} = 247.13 \text{ KNm (Bottom)}$ $= 34.977 \text{ KNm (Top)}$ Moment about y-axis $M_{uy} = 341.684 \text{ KNm (Bottom)}$ $= 164.0233 \text{ KNm (Top)}$ <u>Flexural Design of Column :</u>	
5.	IS 456 :2000	Check For Slenderness Ratio $L_{ex}/D = 4.706 < 12$ $L_{ey}/D = 4.284 < 12$ Minimum Eccentricity ($e_{x \text{ min}}$) $= 21.372 > 20 \text{ mm}$ Minimum Eccentricity ($e_{y \text{ min}}$) $= 20.950 > 20 \text{ mm}$ Moment due to eccentricity $M_{u \text{ ex min}} = P * e_{y \text{ min}} = 44.235 \text{ KNm}$ $M_{u \text{ ey min}} = P * e_{x \text{ min}} = 45.126 \text{ KNm}$ (ETABS result shows the 2.78 % of rebar) Take 2.78% of Rebar in the Column. Gross Area (A_g) = $500*500 = 250000 \text{ mm}^2$	So, Design Short Column

		<u>Along X Direction</u> $d'/D = 56/500 = 0.112$ Assuming reinforcement equal on all side. Take $d'/D = 0.15$ $F_y = 500 \text{ N/mm}^2$ $p/f_{ck} = 2.78/25 = 0.112$ $P_u/f_{ck} bD = 0.3378$ $M_{ux1}/f_{ck} bD^2 = 0.17$ $M_{ux1} = 0.17 \cdot 25 \cdot 500 \cdot 500^2$ $= 531.25 \text{ KNm}$ <u>Along y Direction</u> $d'/b = 56/500 = 0.112 \rightarrow 0.15$ Assuming reinforcement equal on all side. Take $d'/b = 0.15$ $F_y = 500 \text{ N/mm}^2$ $p/f_{ck} = 2.78/25 = 0.112$ $P_u/f_{ck} bD = 0.3378$ $M_{uy1}/f_{ck} D b^2 = 0.17$ $M_{uy1} = 0.17 \cdot 25 \cdot 500 \cdot 500^2$ $= 531.25 \text{ KNm}$ Now , $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$ $= 0.45 \cdot 25 \cdot (97.22/100) \cdot (500 \cdot 500) + 0.75 \cdot 500 \cdot (2.78/100) \cdot (500 \cdot 500)$ $= 5340.562 \text{ KN}$ $\frac{P_u}{P_{uz}} = 0.395$	
	SP : 16		
	SP : 16		
	IS 456: 2000		

6.		For α_n , $0.2 \rightarrow 1$ $0.8 \rightarrow 2$ Interpolating $\alpha_n = 1.325$ $\left(\frac{M_{ux}}{M_{ux1}}\right)^{\alpha_n} + \left(\frac{M_{uy}}{M_{ux1}}\right)^{\alpha_n} \leq 1$ $\left(\frac{247.13}{531.25}\right)^{1.325} + \left(\frac{341.684}{531.25}\right)^{1.325} \leq 1$ $0.9199 \leq 1$ <i>OK</i> Area of steel required = 2.78% of (500*500) = 6950 mm² Provide 16 nos of 25 mm diameter bar Area of steel provided = 7853.98 > 6950 mm² Percentage of steel provided = $\frac{7853.98}{500 * 500} * 100\%$ = 3.141 % (0.8-4%) ok Check for shear Provide 6 - Legged stirrup size of lateral ties (stirrups) = 8mm Area of shear reinforcement = $6 * \frac{\pi * 8^2}{4} =$ 301.592 mm² Diameter of lateral ties = $\frac{1}{4}$ of largest long rebar = $\frac{1}{4} * 25$	
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		=6.25m or 6m , take 8m ok Spacing of Lateral ties • Least of column dimension = 500mm • 16* smaller dia bar = 16*25= 400 mm • 300mm Take 100 mm spacing	
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For Column 600 mm* 600 mm

Design Steps	References	Calculations	Remark
1.	NBC 105: 2025	<u>Member Dimensions</u> Width = 600 mm > 300mm Depth = 600 mm > 300mm Check : 20*16 = 320mm < 600 mm B/D = 1 > 0.45 Assume, Clear Cover = 50 mm Preliminary diameter of L Bar = 16mm Effective Cover = $40 + 16/2 + 8/2 =$ 56 mm	B,D> 300mm OK 20*do < 550mm OK
2.		<u>Material Properties</u> Grade of Concrete (fck) = M25 Grade of Steel (fy) = Fe 500	
3.		<u>Member Properties</u> Length of Column = 2.870 m Depth of beam = 350mm Effective Length Factor : Kx = 0.935 (M3) Ky = 0.920 (M2) Unsupported Length = 2.870 – 0.350= 2.520 mm Effective Length of Column (Lex) =	

4.		Unsupported length * $K_x = 2.520 \times 0.935$ $= 2.356 \text{ mm}$ Similarly , $L_{ey} = 2.3184 \text{ mm}$ <u>Load Calculation</u> Axial load on Column = 2780.106 KN Moment about x-axis $M_{ux} = 399.469 \text{ KNm (Bottom)}$ $= 67.374 \text{ KNm (Top)}$ Moment about y-axis $M_{uy} = 480.977 \text{ KNm (Bottom)}$ $= 2.37 \text{ KNm (Top)}$ <u>Flexural Design of Column :</u> Check For Slenderness Ratio $L_{ex}/D = 3.92 < 12$ $L_{ey}/b = 3.864 < 12$ Minimum Eccentricity ($e_{x \min}$) $= 24.712 > 20 \text{ mm}$ Minimum Eccentricity ($e_{y \min}$) $= 24.63 > 20 \text{ mm}$ Moment due to eccentricity $M_{u \text{ ex min}} = P * e_{y \min} = 68.474 \text{ KNm}$ $M_{u \text{ ey min}} = P * e_{x \min} = 68.701 \text{ KNm}$ Assume 2.02% of Rebar in the Column. Gross Area (A_g) = $600 \times 600 = 360000 \text{ mm}^2$ <u>Along X Direction</u>	
5.	IS 456 :2000		So, Design Short Column

		$d'/D = 56/550 = 0.101$ Assuming reinforcement equal on all side. Take $d'/D = 0.10$ $F_y = 500 \text{ N/mm}^2$ $p/f_{ck} = 2.02/25 = 0.08$ $P_u/f_{ck} bD = 0.308$ $M_{ux1}/f_{ck} bD^2 = 0.135$ $M_{ux1} = 0.140 \cdot 25 \cdot 600 \cdot 600^2$ $= 756 \text{ KNm}$ <u>Along y Direction</u> $d'/b = 56/500 = 0.112$ Assuming reinforcement equal on all side. Take $d'/b = 0.10$ $F_y = 500 \text{ N/mm}^2$ $p/f_{ck} = /25 = 0.08$ $P_u/f_{ck} bD = 0.308$ $M_{uy1}/f_{ck} bD^2 = 0.135$ $M_{uy1} = 0.135 \cdot 25 \cdot 600 \cdot 600^2$ $= 756 \text{ KNm}$ Now , $P_{uz} = 0.45 f_{ck} A_c + 0.75 f_y A_{sc}$ $= 0.45 \cdot 25 \cdot (100 - 2.02/100) \cdot (600 \cdot 600) + 0.75 \cdot 500 \cdot (2.02/100) \cdot (600 \cdot 600)$ $= 6695.190 \text{ KN}$ $\frac{P_u}{P_{uz}} = 0.415$ For α_n ,	
	SP : 16		
	SP : 16		
	IS 456: 2000		

6.		0.2 → 1 0.8 → 2 Interpolating $\alpha_n = 1.358$ $\left(\frac{M_{ux}}{M_{ux1}}\right)^{\alpha_n} + \left(\frac{M_{uy}}{M_{ux1}}\right)^{\alpha_n} \leq 1$ 0.961 ≤ 1 <i>OK</i> Area of steel required = 2.02% of (600*600) = 7272 mm² Provide 16 nos of 25 mm diameter bar Area of steel provided = 7853 > 7272mm² Percentage of steel provided = $\frac{7853}{600 * 600} * 100\%$ = 2.181% (0.8-6%) ok Check for shear Provide 4- Legged stirrup size of lateral ties (stirrups) =8mm Area of shear reinforcement = $8 * \frac{\pi * 8^2}{4} =$ 402.12 mm² Diameter of lateral ties = $\frac{1}{4}$ of largest long rebar = $\frac{1}{4} * 25$ =6.25mm or 6mm , take 8mm ok Spacing of Lateral ties • Least of column dimension = 600 mm • 16* smaller dia bar = 16*25 = 400 mm	
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		• 300mm Take 100 mm spacing	
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5.3 Design of Staircase

A staircase provides safe and convenient vertical access between different floors of a building. The configuration of a staircase depends on architectural planning, available space, and functional requirements.

In the present residential building, an open well staircase is adopted. This type of staircase consists of two or more flights arranged around an open central space (well). The flights are separated by landings and typically run parallel to each other with a clear opening between them. Open well staircases are commonly used in residential buildings due to their aesthetic appeal, improved lighting and ventilation, and ease of movement.

Structurally, the stair slab is designed to resist dead loads from self-weight and finishes, along with live loads due to occupancy. The design is carried out by considering effective span, support conditions, load distribution, and slab thickness.

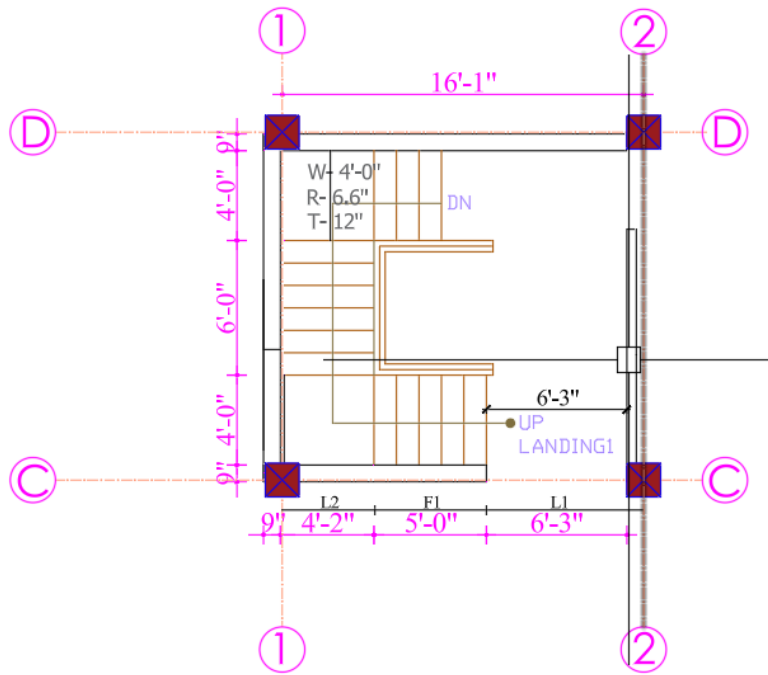
Dimensional Requirements (As per NBC 201 – Residential Buildings)

For residential buildings, NBC specifies:

- **Minimum clear width of staircase:** 900 mm
- **Maximum riser height:** 190 mm
- **Minimum tread depth:** 250 mm
- **Minimum headroom clearance:** 2.1 m
- **Landing width:** Not less than width of stair

The comfort relationship is generally maintained as:

$$2R + T = 600 \text{ to } 650 \text{ mm}$$



Design of openwell staircase

Floor to floor height = 2.870 m

Riser(R) = 165 mm

Tread(T) = 300 mm

Landing(L1) = 1905 mm

Flight(f1) = 1500 mm

Landing (L2) = 1200 mm

Flight(F2) = 2800 mm

Landing(L3) = 1200 mm

Flight(F3) = 900 mm

Landing(L4) = 2475 mm

Number of riser in flight 1(N1) = 6

Number of riser in flight 2 (N2) = 7

Number of riser in flight 3 (N3) = 4

Width of stair = 1200 mm

Imposed load = 3 KN/M²

Floor finishes = 1KN/M²

Brickwall width (B) = 230 mm

Unit wt. of concrete = 25KN/M²

Grade of steel (fy) = 500N/mm²

Grade of concrete (fck) = 25N/mm²

Design of Landing 1 and Flight F1

Step1: calculation of effective span

$$\begin{aligned} \text{Effective span}(l_{\text{eff}}) &= \{(N1-1) * T\} + L1 + (L2/2) + B/2 \\ &= \{(6-1) * 300\} + 1905 + 600 + 115 \\ &= 4120 \text{ mm} \end{aligned}$$

Step2: calculation of effective depth

Clear cover(c) = 15mm **IS 456: 2000, Table 16**

Adopt diameter of bar (φ) = 12 mm

$l_{\text{eff}}/d = 30$, (assuming 30-40)

Effective depth(d) = ($l_{\text{eff}}/30$)

$$= 137.33 \text{ mm}$$

$$\begin{aligned}\text{Overall depth (D)} &= d + (\phi/2) + cc \\ &= 137.33 + 6 + 15 \\ &= 158.33\text{mm} = 165\text{mm}\end{aligned}$$

Provided eff. depth = 140 mm

Step 3: calculation of load

(i) Load on going on projected plan area

Live load = 3 kN/m

Floor finish = 1 kN/m

Wt. of waist slab = $(\gamma \cdot D \cdot (\sqrt{R^2 + T^2})/T)$, length of sloping slab = $(\sqrt{R^2 + T^2})$
= 4.71 kN/m

Wt. of steps = $\{(0.5 \cdot R \cdot T) \cdot \gamma\}/T$
= 2.06 kN/m

Total load = 10.77 kN/m

Factored load (W) = $1.5 \cdot 10.77 = 16.155$ kN/m

(ii) Load on landing

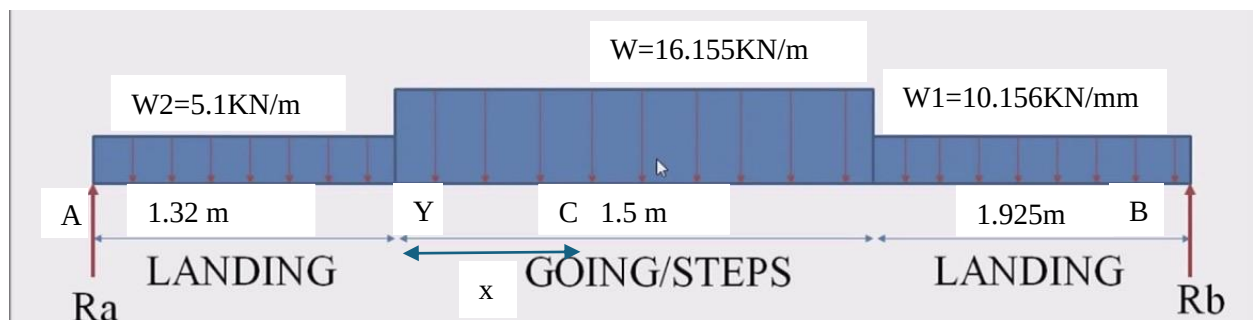
Live load = 3 kN/m

Floor finish = 1 kN/m

Self wt. of slab = $\gamma \cdot D \cdot 1$
= $25 \cdot 0.165 \cdot 1$
= 4.125 kN/m

Total load = 8.125 kN/m

Factored load (w1) = 10.156 kN/m



Step 4: calculation of maximum bending moment and maximum shear force

Taking moment of all forces about point B.

$$R_a \cdot 4.74 = \{5.1 \cdot 1.32 \cdot (1.32/2 + 1.5 + 1.925)\} + \{16.155 \cdot 1.5 \cdot (1.5/2 + 1.925)\} + \{10.156 \cdot 1.925 \cdot 1.925/2\}$$

$$R_a = 23.447 \text{ kN}$$

$$R_a + R_b = 5.1 \cdot 1.32 + 16.155 \cdot 1.5 + 10.156 \cdot 1.925$$

$$R_b = 27.06 \text{ kN}$$

Zero shear force point (at a distance X from point Y)

$$\begin{aligned}X &= \{23.477 - (5.1 \cdot 1.32)\} / 16.155 \\ &= 1.04 \text{ m}\end{aligned}$$

Taking moment about point C

$$\Sigma M_c = \{(23.477 \cdot (1.32 + 1.04)) - (5.1 \cdot 1.32) \cdot (1.32/2 + 1.04) - (16.155 \cdot 1.04) \cdot 1.04/2\}$$

$$M_{\max} = 35.22 \text{ kN}$$

Step 5: Check for depth against bending moment

$$M_{\max} = 0.133 \cdot f_{ck} \cdot b \cdot d_{\text{req}}^2$$

$$35.22 \cdot 10^6 = 0.133 \cdot 25 \cdot 1000 \cdot d_{\text{req}}^2$$

$$d_{\text{req}} = 102.92 \text{ mm}, (d_{\text{req}} < d_{\text{prov}}) \text{ .Hence ok}$$

Step6 : Designing the reinforcement

$$M_{max} = 0.87 \cdot F_y \cdot A_{st} \cdot \left(d_{prov} - \frac{F_y \cdot A_{st}}{F_{ck} \cdot b} \right)$$

$$35.22 \cdot 10^6 = 0.87 \cdot 500 \cdot A_{st} \cdot \left(140 - \frac{500 \cdot A_{st}}{25 \cdot 1000} \right)$$

$$A_{st} = 636.134 \text{ mm}^2$$

$$(A_{st})_{min} = 0.12\% \text{ of } bD$$

$$= 0.12\% \cdot 1000 \cdot 165$$

$$= 192 \text{ mm}^2$$

$$\text{Spacing} = \frac{1000 \cdot 3.142 \cdot 12^2 / 4}{636.134}$$

$$= 177.788 \text{ mm}$$

Provide 12mm ϕ bar @ 150mm c/c

$$(A_{st})_{prov} = \frac{1000 \cdot 3.142 \cdot 12^2 / 4}{150} = 753.982 \text{ mm}^2$$

$$\% \text{ of steel prov.} = \frac{753.982}{1000 \cdot 140} \cdot 100\%$$

$$= 0.54\%$$

Step7: Check for shear force

For M25 concrete

Shear strength(T_c) = 0.28 N/mm²

$$\text{Nominal shear stress}(T_v) = \frac{V_{max}}{bd} \cdot 10^3$$

$$= \frac{27.06 \cdot 10^3}{1000 \cdot 140}$$

$$= 0.193 \text{ N/mm}^2$$

(T_c)_{max} = 3.1 N/mm², ok

Step8: Check for deflection

$$\left(\frac{l}{d}\right)_{all} = \alpha \cdot \beta \cdot \gamma \cdot \delta \cdot \lambda$$

Basic value (α) = 26 IS 456:2000 Clause 23.2.1(a)

Modification factor (β) = 1 IS 456:2000 Clause 23.2.1(b)

For compression reinforcement (δ) = 1 IS 456:2000 Clause 23.2.1(d) fig 5

Reduction factor for flanged beam (λ) = 1 IS 456:2000 Clause 23.2.1(e) fig 6

For calculating the factor (γ)

$$\text{Steel stress of service load}(f_s) = 0.58 \cdot f_y \cdot \frac{\text{Area of steel required}}{\text{Area of steel provided}}$$

$$= 0.58 \cdot 500 \cdot \frac{636.134}{753.982}$$

$$= 244.672 \text{ N/mm}^2$$

$$A_{st} \% = \frac{\text{Area of steel provided}}{bd} \cdot 100\%$$

$$= \frac{753.982}{1000 \cdot 140} \cdot 100\%$$

$$= 0.54\%$$

From fig .4 of IS 456:2000 (γ) = 1.14

Now,

$$\alpha \cdot \beta \cdot \gamma \cdot \delta \cdot \lambda = 26 \cdot 1 \cdot 1.18 \cdot 1 \cdot 1 = 30.68 \text{ mm}$$

$$\left(\frac{l}{d}\right)_{pro} = 4120 / 140$$

$$= 24.42 \text{ mm} < \left(\frac{l}{d}\right)_{all}, \text{ ok}$$

Step9: Development length

$$L_d = \frac{\phi \cdot 0.87 \cdot f_y}{4 \cdot T_{bd}}$$

$$= \frac{12 \cdot 0.87 \cdot 500}{4 \cdot 1.2 \cdot 1.6}$$

$$= 679.687 \text{ mm}$$

Step10: Distribution bar reinforcement detailing

Adopt distribution bar dia.=8mm

$$\text{Spacing} = \frac{1000 \times 50.26}{192} \\ = 261.77 \text{ mm}^2$$

Provide a distribution bar of 8mm dia.@200mm c/c

Design of landing 2, flight 2 and landing 3

Step1: calculation of effective span

$$\begin{aligned} \text{Effective span}(l_{\text{eff}}) &= \{(N2-1) * T\} + (L2/2) + (L3/2) \\ &= \{(7-1) * 300\} + 1905/2 + 600 \\ &= 3000 \text{ mm} \end{aligned}$$

Step2: calculation of effective depth

Clear cover(c)= 15mm **IS 456: 2000, Table 16**

Adopt diameter of bar (ϕ) = 12 mm

$l_{\text{eff}}/d = 30$, (assuming 30-40)

$$\begin{aligned} \text{Effective depth}(d) &= (l_{\text{eff}}/30) \\ &= 100 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overall depth}(D) &= d + (\phi/2) + cc \\ &= 100 + 6 + 15 \\ &= 121 \text{ mm} = 125 \text{ mm} \end{aligned}$$

Provided eff.depth=100 mm

Step3: calculation of load

(i) Load on going on projected plan area

Live load = 3KN/m

Floor finish = 1KN/m

$$\begin{aligned} \text{Wt. of waist slab} &= (\gamma * D * (\sqrt{R^2 + T^2}) / T, \text{ length of sloping slab} = (\sqrt{R^2 + T^2}) \\ &= 3.57 \text{ KN/m} \end{aligned}$$

$$\begin{aligned} \text{Wt. of steps} &= \{(0.5 * R * T * \gamma) / T\} \\ &= 2.06 \text{ KN/m} \end{aligned}$$

Total load = 9.53 KN/m

$$\text{Factored load}(W) = 1.5 * 9.53 = 14.155 \text{ KN/m}$$

(ii) Load on landing

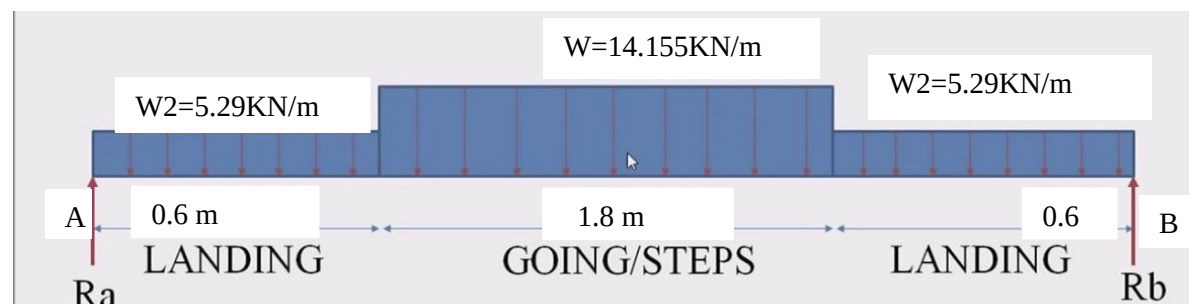
Live load = 3KN/m

Floor finish = 1KN/m

$$\begin{aligned} \text{Self wt. of slab} &= \gamma * D * 1 \\ &= 25 * 0.125 * 1 \\ &= 3.125 \text{ KN/m} \end{aligned}$$

Total load = 7.125 KN/M

$$\text{Factored load}(w1) = 10.688 \text{ KN/m}$$



Step4: calculation of maximum bending moment and maximum shear force

$$\text{Total load}(W)=5.29*0.6+14.155*1.8+5.29*0.6=31.827\text{KN}$$

$$V_{\max} = \frac{W \cdot L}{2}$$

$$= 47.74 \text{ KN}$$

Moment is maximum at center point of loading diagram shown above.

$$M_{\max} = 14.328 \text{ KN-m}$$

Step5: Check for depth against bending moment

$$M_{\max} = 0.133 \cdot f_{ck} \cdot b \cdot d_{\text{req}}^2$$

$$14.328 \cdot 10^6 = 0.133 \cdot 25 \cdot 1000 \cdot d_{\text{req}}^2$$

$$d_{\text{req}} = 65.64 \text{ mm}, (d_{\text{req}} < d_{\text{prov}}) \text{ .Hence ok}$$

Step6 : Designing the reinforcement

$$M_{\max} = 0.87 \cdot F_y \cdot A_{st} \cdot \left(d_{\text{prov}} - \frac{F_y \cdot A_{st}}{F_{ck} \cdot b} \right)$$

$$14.328 \cdot 10^6 = 0.87 \cdot 500 \cdot A_{st} \cdot \left(100 - \frac{500 \cdot A_{st}}{25 \cdot 1000} \right)$$

$$A_{st} = 354.515 \text{ mm}^2$$

$$(A_{st})_{\min} = 0.12\% \text{ of } bD$$

$$= 0.12\% \cdot 1000 \cdot 125$$

$$= 150 \text{ mm}^2$$

$$\text{Spacing} = \frac{1000 \cdot 3.142 \cdot 12^2 / 4}{354.515}$$

$$= 319.01 \text{ mm}$$

Provide 12mm ϕ bar @ 150mm c/c

$$(A_{st})_{\text{prov}} = \frac{1000 \cdot 3.142 \cdot 12^2 / 4}{150} = 753.982 \text{ mm}^2$$

$$\% \text{ of steel prov.} = \frac{753.982}{1000 \cdot 100} \cdot 100\%$$

$$= 0.75\%$$

Step7: Check for shear force

For M25 concrete

$$\text{Shear strength}(T_c) = 0.28 \text{ N/mm}^2$$

$$\text{Nominal shear stress}(T_v) = \frac{V_{\max}}{\frac{bd}{1000 \cdot 100}} \cdot 10^3$$

$$= \frac{47.74 \cdot 10^3}{1000 \cdot 100}$$

$$= 0.477 \text{ N/mm}^2$$

$$(T_c)_{\max} = 3.1 \text{ N/mm}^2$$

$$T_v > T_c, T_v < T_{c\max}, \text{ ok}$$

Step8: Check for deflection

$$\left(\frac{l}{d} \right)_{\text{all}} = \alpha \cdot \beta \cdot \gamma \cdot \delta \cdot \lambda$$

$$\text{Basic value } (\alpha) = 26 \quad \text{IS 456:2000 Clause 23.2.1(a)}$$

$$\text{Modification factor } (\beta) = 1 \quad \text{IS 456:2000 Clause 23.2.1(b)}$$

$$\text{For compression reinforcement } (\delta) = 1 \quad \text{IS 456:2000 Clause 23.2.1(d) fig 5}$$

$$\text{Reduction factor for flanged beam } (\lambda) = 1 \quad \text{IS 456:2000 Clause 23.2.1(e) fig 6}$$

For calculating the factor (γ)

$$\text{Steel stress of service load}(f_s) = 0.58 \cdot f_y \cdot \frac{\text{Area of steel required}}{\text{Area of steel provided}}$$

$$= 0.58 \cdot 500 \cdot \frac{354.515}{753.982}$$

$$= 136.355 \text{ N/mm}^2$$

$$A_{st} \% = \frac{\text{Area of steel provided}}{bd} \cdot 100\%$$

$$= \frac{753.982}{1000 \cdot 125} \cdot 100\%$$

$$= 0.6\%$$

$$\text{From fig .4 of IS 456:2000 } (\gamma) = 1.87$$

Now,

$\alpha \cdot \beta \cdot \gamma \cdot \delta \cdot \lambda = 26 \cdot 1 \cdot 1.87 \cdot 1 \cdot 1 = 48.62 \text{ mm}$, each flight is connected to landing slabs.

$$\left(\frac{l}{d}\right)_{\text{pro}} = 3000 / 125 \\ = 24 \text{ mm} < \left(\frac{l}{d}\right)_{\text{all}} , \text{ ok}$$

Step9: Development length

$$L_d = \frac{\phi \cdot 0.87 \cdot f_y}{\frac{4 \cdot T_b d}{12 \cdot 0.87 \cdot 500}} \\ = \frac{4 \cdot 1.2 \cdot 1.6}{4 \cdot 1.2 \cdot 1.6} \\ = 679.687 \text{ mm}$$

Step10: Distribution bar reinforcement detailing

Adopt distribution bar dia.=8mm

$$\text{Spacing} = \frac{1000 \cdot 50.26}{150} \\ = 335.06 \text{ mm}^2$$

Provide a distribution bar of 8mm dia.@200mm c/c

5.4 Design of Basement wall

A basement wall is constructed to retain surrounding soil and to prevent moisture ingress into the building. Since the wall is supported by a mat foundation, overall stability is ensured, and the design primarily focuses on the safe performance of the vertical stem.

Basement walls form the exterior boundary of underground structures such as basements, tunnels, and earth-retained spaces. These walls must resist lateral earth pressure along with

any additional loads arising from surcharge or other external actions. Structurally, basement walls behave as vertical slabs supported by the floor system at the basement and upper levels. The axial forces from floor framing are either resisted by shear walls or balanced by lateral earth pressure acting from the opposite side of the structure.

Although basement walls function as vertically supported slabs when connected to floor systems, they should be designed as cantilever elements during early construction stages when the upper floor has not yet been completed.

1. Material Properties:

Grade of concrete(f_{ck}): 35 MPa

Grade of steel(f_y)=500 MPa

Modulus of elasticity of steel (E_s) = 200000 MPa

Unit weight of soil(γ) = 19 KN/m³

Angle of Internal Friction(ϕ)= 30°

Surcharge due to vehicular movement(w)= 20 KN/m²

$k_a = (1 - \sin\phi / 1 + \sin\phi) = 0.33$

2. Basement Wall Data:

Height of basement wall= 3.048m

Width of basement wall= 1m

Clear Cover(cc)= 50mm

Effective cover(d')= 72mm [cc+ dia of shear bar+ dia of main bar/2]

3. Load Calculation:

Factored lateral load due to surcharge(P_1)= $k_a * w * h = 41$ KN/m

Factored lateral load due to soil pressure(P_2)= $\frac{1}{2} * k_a * \gamma * h^2 = 78$ KN/m

4. Flexural Design of Basement Wall:

Ultimate Moment(M_u)= 141.732 KN-m

For Fe500, BM= $0.133 f_{ck} b d^2$

On solving,

$d = 175$ mm

$D = 250$ mm

Then limiting moment (M_{limit}) (AS per IS 456: 2000 cl G-1.1 c) = $0.36 F_{ck} b X_m$
($d - 0.416 X_m$)

$0.46d$

$X_m =$

$0.46 * 175 = 80.5$

=

= 103.66

KNm

Since $M_u > M_{u \text{ limit}}$, the section is Doubly reinforced section.

5. Calculation of main Reinforcement :

Calculation Of tension Reinforcement

$$M_{u \text{ limit}} = 0.87 f_y A_{t1} \left(d - \frac{f_y A_{t1}}{b f_{ck}} \right)$$

$$A_{t1} = 1560.495 \text{ mm}^2$$

Again,

$$(\text{BM max} - \mu \text{ limit}) = 0.87 f_y A_{t2} \left(d - \frac{f_y A_{t2}}{b f_{ck}} \right)$$

$$A_{t2} = 522.402 \text{ mm}^2$$

$$A_{t \text{ required}} = A_{t1} + A_{t2} = 2028.897 \text{ mm}^2$$

$$A_{t \text{ min}} = 0.12\% \times bD = 300 \text{ mm}^2, \text{ Ok}$$

Provide 16 mm diameter bar

$$\text{Spacing Required} = \frac{A_{1\phi} \times 1000}{A_{t \text{ req}}} = 100 \text{ mm}$$

Provide 95 mm spacing

$$A_{t \text{ Provided}} = \frac{A_{1\phi} \times 1000}{\text{Spacing Provided}} = 2116.42 \text{ mm}^2 > A_{t \text{ required}}$$

Calculation of compression reinforcement:

$$\text{Strain at the level of compression reinforcement (Esc)} = 0.0035 (1 - d'/x_m) = 0.000369$$

$$(\text{BM max} - \mu \text{ limit}) = 0.36 F_{ck} b X_u + (F_{sc} - F_{ck}) A_{sc} (d - d')$$

$$\text{Assuming } A_{sc} \text{ required} = 522.402 \text{ mm}^2$$

Provide 12mm diameter bar.

$$\text{Spacing required} = \frac{A_{1\phi} \times 1000}{A_{sc \text{ req}}} = 216.494 \text{ mm}$$

Provide 180mm spacing

$$A_{t \text{ Provided}} = \frac{A_{1\phi} \times 1000}{\text{Spacing Provided}} = 628 \text{ mm}^2 > A_{t \text{ required}}$$

Design of Beam

1. Design of Primary Beam:

Beam ID : B11 Floor : Third Floor

Concrete Grade = M25

Steel Grade = Fe500

1.1. Design of Longitudinal bars for moment:

Ref	Step	Calculation	Output/ Remarks
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IS 456:2000 Cl.6.2.1 IS 13920:2016 Cl.6.2.1	1. 2. 3.	Known Data Overall Depth of Beam, D = 500mm Width of Beam, B = 350mm Effective Cover = 60mm Diameter of the bar, ϕ = 16 mm Considering bars in single layer Effective depth, d = 500 – 60 = 440 mm Check for member size $\frac{B}{D} = \frac{350}{500} = 0.7$ Clear Length, L = 4902 – 500/2 – 500/2 = 4402 m L/D = 4402/500 = 8.804 Minimum Area Required ($A_{st,min}$) $= \frac{0.85}{f_y} * b * d$ $= \frac{0.85}{500} * 350 * 440$ $= 261.8 \text{ mm}^2$ $= 0.24 \frac{\sqrt{f_{ck}}}{f_y} * b * d$ $= 0.24 \frac{\sqrt{25}}{500} * 350 * 500$ $= 420 \text{ mm}^2$	>0.3 (OK) >4 (OK) Adopt Max= 420 mm²
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	4.	Maximum Area Required Tension Steel ($A_{st,max}$) = $0.04 \cdot b \cdot d$ $= 0.04 \cdot 350 \cdot 500$ $= 7000 \text{ mm}^2$ Compression Steel ($A_{sc,max}$) = $0.04 \cdot b \cdot d$ $=$ $0.04 \cdot 350 \cdot 500$ $= 7000 \text{ mm}^2$	
	5.	Limiting Value of depth of neutral axis $\frac{x_{u,max}}{d} = \frac{0.0035}{0.0055 + \frac{f_y}{1.115 \cdot E_s}}$ $= 0.46$	
	6.	Limiting Moment (M_{ulim}) $M_{ulim} = 0.133 \cdot f_{ck} \cdot b \cdot d^2$ $= 0.133 \cdot 25 \cdot 350 \cdot 500^2$ $= 290.93 \text{ KNm}$	
	7.	Position: Top End I Governing Combination=Envelope Factored Moment, $M_u = 164.166 \text{ KNm}$ Torsional Moment, $T_u = 11.67 \text{ KNm}$ Moment due to torsion (M_t) = $\frac{T_u \cdot (1 + \frac{D}{b})}{1.7}$ $= 16.67 \text{ KNm}$ Required Bending Moment(M) = $M_u + M_t$ $= 164.166 + 16.67$ $= 180.836 \text{ KNm}$ Since $M < M_{ulim}$ design as singly reinforced $A_{st} = 0.5 \frac{f_{ck}}{f_y} \left(1 - \sqrt{1 - 4.6 \frac{M_u}{f_{ck} \cdot b \cdot d^2}} \right) \cdot b \cdot d$ $= 834.81 \text{ mm}^2 \text{ (Top)}$	$> A_{stmin}$ & $< A_{stmax}$
	8.	Percentage of tension steel (%) = 1% Also, Area of Compression Steel(A_{sc}) = 420 mm^2 Percentage of compression steel(%) = 0.5% Position: Top Middle	$A_{sc} =$ 50% of A_{st} or A_{stmin} whichever is greater

		Governing Combination=Envelope Factored Moment, $M_u = 66.36$ KNm Torsional Moment, $T_u = 11.67$ KNm Moment due to torsion (M_t) = $\frac{T_u * (1 + \frac{D}{b})}{1.7}$ = 16.67 KNm Required Bending Moment(M) = $M_u + M_t$ = 66.36+16.67 = 83.03 KNm Since $M < M_{ulim}$ design as singly reinforced $A_{st} = 0.5 \frac{f_{ck}}{f_y} (1 - \sqrt{1 - 4.6 \frac{M_u}{f_{ck} * b * d^2}}) * bd$ = 316.72 mm² (Top) Percentage of tension steel (%) = 1% Also, Area of Compression Steel(A_{sc}) = 420 mm² Percentage of compression steel(%) = 0.5% Position: Top End J Governing Combination=Envelope Factored Moment, $M_u = 279.98$ KNm Torsional Moment, $T_u = 11.67$ KNm Moment due to torsion (M_t) = $\frac{T_u * (1 + \frac{D}{b})}{1.7}$ = 16.67 KNm Required Bending Moment(M) = $M_u + M_t$ = 279.98+16.67 = 296.65 KNm Since $M > M_{ulim}$ design as doubly reinforced $A_{st1} = 0.5 \frac{f_{ck}}{f_y} (1 - \sqrt{1 - 4.6 \frac{M_{ulim}}{f_{ck} * b * d^2}}) * bd$ = 1649.07mm² For A_{st2} $M_u - M_{ulim} = 0.87 f_y A_{st2} (d - d')$ Gives, $A_{st2} = 34.603$ mm² Total $A_{st} = 1683.67$ mm² Percentage of tension steel (%) = 1.02% Also, Area of Compression Steel(A_{sc}) = 841.835	>A_{stmin}& <A_{stmax} $A_{sc} =$ 50% of A_{st} or A_{stmin} whichever is greater $A_{sc} =$ 50% of A_{st} or A_{stmin} whichever
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	10.	mm² Percentage of compression steel(%) = 0.51% Position: Bottom End I Governing Combination=Envelope Factored Moment, M_u = 72.45 KNm Torsional Moment, T_u = 11.67 KNm Moment due to torsion (M_t) = $\frac{T_u * (1 + \frac{D}{b})}{1.7}$ = 16.67 KNm Required Bending Moment(M) = M_u+M_t = 72.45+16.67 = 89.12 KNm Since M<M_{ulim} design as singly reinforced $A_{st} = 0.5 \frac{f_{ck}}{f_y} (1 - \sqrt{1 - 4.6 \frac{M_u}{f_{ck} * b * d^2}}) * bd$ = 347.03 mm² (bot)	is greater >A_{stmin}& <A_{stmax} Asc = 50% of A_{st} or A_{stmin} whichever is greater
	11.	Percentage of tension steel (%) = 1% Also, Area of Compression Steel(A_{sc}) = 420 mm² Percentage of compression steel(%) = 0.5% Position: Bottom Middle Governing Combination=Envelope Factored Moment, M_u = 147.98 KNm Torsional Moment, T_u = 11.67 KNm Moment due to torsion (M_t) = $\frac{T_u * (1 + \frac{D}{b})}{1.7}$ = 16.67 KNm Required Bending Moment(M) = M_u+M_t = 147.98+16.67 = 164.65 KNm Since M<M_{ulim} design as singly reinforced $A_{st} = 0.5 \frac{f_{ck}}{f_y} (1 - \sqrt{1 - 4.6 \frac{M_u}{f_{ck} * b * d^2}}) * bd$ = 743.96 mm²	>A_{stmin}& <A_{stmax} Asc = 50% of A_{st} or A_{stmin} whichever is greater
	12.	Percentage of tension steel (%) = 1% Also, Area of Compression Steel(A_{sc}) = 420 mm²	A_{st} or A_{stmin} whichever is greater

		Percentage of compression steel(%) = 0.5% Position: Bottom End J Governing Combination=Envelope Factored Moment, $M_u = 190.05$ KNm Torsional Moment, $T_u = 11.67$ KNm Moment due to torsion (M_t) = $\frac{T_u * (1 + \frac{D}{b})}{1.7}$ = 16.67 KNm Required Bending Moment(M) = $M_u + M_t$ = 190.05 + 16.67 = 206.72 KNm Since $M < M_{ulim}$ design as singly reinforced $A_{st} = 0.5 \frac{f_{ck}}{f_y} (1 - \sqrt{1 - 4.6 \frac{M_u}{f_{ck} * b * d^2}}) * b d$ = 985.145 mm² Percentage of tension steel (%) = 1% Also, Area of Compression Steel(A_{sc}) = 492.57 mm² Percentage of compression steel(%) = 0.5%	$> A_{stmin}$ & $< A_{stmax}$ $A_{sc} =$ 50% of A_{st} or A_{stmin} whichever is greater
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Longitudinal Detailing :

Position	Manually calculated Area of Steel (mm ²)	Bars	Area Provided (mm ²)	Provided % Ast
Left (Top)	834.81	3-20 ϕ	942.477	0.53
Left (Bottom)	347.03	2-20 ϕ	628.31	0.35
Mid (Top)	316.72	2-20 ϕ	628.31	0.35
Mid (Bottom)	743.96	3-20 ϕ	942.477	0.53
Right (Top)	1683.67 in Tension , 841.835 in Compression (Doubly)	6-20 ϕ (Tension), 3-20 ϕ (Compression)	1885.71(Tension) 942.477 (Compression)	1.08(T), 0.53(C)
Right (Bottom)	985.145	4-20 ϕ	1256.63	0.71

Provision of Transverse bars for Shear Reinforcement

Since $d > 450$ mm, Additional reinforcement is required.
Side face reinforcement shall be provided along the two faces.
Therefore, Provide 20mm dia 2 nos of side reinforcement.

5.5 DESIGN OF MAT FOOTING

A mat foundation, also known as a raft foundation, is a reinforced concrete slab of large area provided to support multiple columns and walls of a structure. It spreads the total structural load over the entire foundation area and transfers it safely to the underlying soil. In residential buildings, mat foundations are commonly adopted where soil bearing capacity is low or where individual footings are likely to overlap.

The net upward soil pressure acting on the mat is calculated by dividing the total vertical load by the plan area of the raft. The soil pressure distribution may be assumed as uniform or linearly varying depending on load eccentricity and soil characteristics. The calculated soil pressure should be less than the allowable bearing capacity of the soil.

C

1. Input parameters

Grade of concrete used(f_{ck})=30 N/mm²

Grade of steel (f_y)=500 N/mm²

Bearing capacity of soil used (SBC)= 100 KN/mm²

Grid spacing along x-axis

Along 1-2= 4.902m

Along 2-3= 4.775m

Grid spacing along y-axis

Along A-B= 3.606m

Along B-C= 3.149m

Along C-D= 4.724 m

Along D-E= 3.20 m

Projection from all four sides=1m

Total length

Total length of footing(L_f)=11.677m

Total width of footing(B_f)= 16.679m

2. Calculation of eccentricity

Node	F_z (KN)
A1	333.676
A2	696.391
A3	434.124
B1	487.397

B2	2061.658
B3	514.994
C1	675.022
C2	2831.736
C3	564.406
D1	603.840
D2	2773.497
D3	582.967
E1	391.035
E2	511.217
E3	427.692

Total vertical load(P_U)= 13889.652 KN

About 1-1

Xbar=

$$\frac{(511.217+2773.497+2831.736+2061.658+696.391)*4.902+(427.692+582.967+564.406+514.994+434.124)*4.902+4.775)}{13889.652}$$

= 4.891 m

$$Ybar = \frac{(487.397+2061.658+514.994)*3.606+(675.022+2831.736+564.406)*(3.606+3.149)+(603.840+2773.497+582.967)*(4.724+3.149+3.606)+(391.035+511.217+427.692)*14.679}{13889.652}$$

= 7.453 m

$$E_x = \frac{9.667}{2} - 4.891 = -0.0525m$$

$$E_y = \frac{14.679}{2} - 7.434 = -0.114m$$

1. Net soil pressure(σ)= $P/A \pm M_x*Y/I_x \pm M_y*X/I_y$

$$I_x = \frac{BD^3}{12} = 4515.02 m^4$$

$$I_Y = \frac{B^3 D}{12} = 2213.006 \text{ m}^4$$

Live load = 0.00 KN/m²

$$M_X = P_U \cdot E_Y = 1576.475 \text{ KN-m}$$

$$M_Y = P_U \cdot E_X = 729.206 \text{ KN-m}$$

$$P/A = 71.316 \text{ KN/m}^2$$

Corner column(500*500)

S.N	Grid	X	Y	σ(KN)	
3	A3	4.838	-7.34	70.347	
	B3	4.838	-3.734	71.606	
	C3	4.838	-0.585	91.607	
	D3	4.838	4.139	93.263	
	E3	4.838	7.339	94.386	
1	A1	-4.838	7.34	48.246	
	C1	-4.838	0.585	69.517	
	D1	-4.838	-4.139	71.167	
	E1	-4.838	-7.339	53.394	

In a y direction the raft is divided into 3 strips equivalent beams. Beam 1-1 with 3.45 m width and soil pressure 82.366 KN/m². Beam 2-2 with 4.84 m width and soil pressure 66.593 KN/M²
Beam 3-3 with 3.39 m width and soil pressure 50.82 KN/m².

For beam 1-1

$$BM_{\max} = wl^2/10$$

$$= 82.366 \cdot 4.902^2/10$$

$$= 183.809 \text{ KN-m}$$

2-2

$$BM_{\max} = 148.610 \text{ KN-m}$$

$$BM_{\max} \text{ 3-3} = 113.410 \text{ KN-m}$$

In a x direction along grid A-A, width 2.80 m

Soil pressure = 73.89 KN/m²

BM_{max} = 221.943 KNm

Beam E-E with width 1.6 m

BM_{max} = 178.107 KNm

3 Calculation of depth

Factored max.BM = 221.943 KNm

$$BM = 0.133 f_{ck} b d^2$$

$$221.943 \times 10^6 = 0.133 \times 30 \times 1000 \times d^2$$

$$d = 235.849 \text{ mm (minimum)}$$

Check for two way shear

$$\tau_c = k_s \times 0.25 \sqrt{f_{ck}}$$

$$K_s = 0.5 + \beta$$

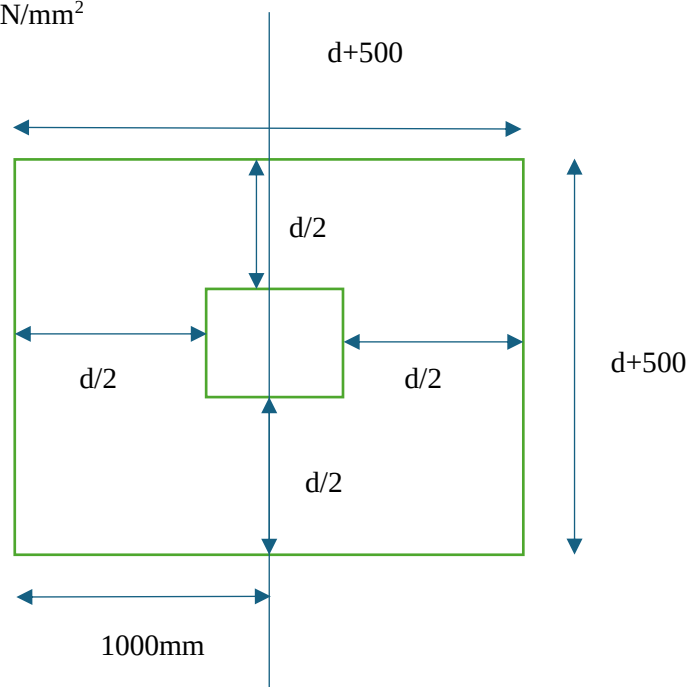
$$B = 500/500 = 1, k_s = 1.5 > 1; k_s = 1$$

$$\tau_c = 0.25 \sqrt{30} = 1.369 \text{ N/mm}^2$$

Corner column

$$\tau_v = \frac{V_u}{P_o d}$$
$$= \frac{391.035 \times 10^3}{4(d+500) \times d} \leq \tau_c$$

$$d \geq 115.935 \text{ mm}$$



D-2 COLUMN

$$P_u = 2773.497 \text{ KN}$$

$$\tau_v = V_u / P_o d$$

$$= \frac{2773.49 \times 10^3}{4(d+600) \times d} \leq \tau_c$$

$$d \geq 472.321 \text{ mm}$$

C-2 COLUMN

$$P_u = 2831.736 \text{ KN}$$

$$\tau_v = V_u / P_o d$$

$$= \frac{2831.736 \times 10^3}{4(d+600) \times d} \leq \tau_c$$

$$d \geq 479.178 \text{ mm}$$

Let us adopt the depth of slab 550 mm and clear cover be 50 mm.

$$\text{Overall depth}(D) = 600 \text{ mm}$$

Calculation of reinforcement

$$BM_{\max} = 0.87 f_y A_{st} \left(d - \frac{f_y A_{st}}{f_{ck} b} \right)$$

$$221.943 \times 10^6 = 0.87 \times 500 \times A_{st} \left(550 - \frac{500 \times A_{st}}{30 \times 1000} \right)$$

$$A_t = 955.316 \text{ mm}^2$$

$$A_{t \min} = 0.12 \% = 720 \text{ mm}^2$$

$$\text{Spacing} = \frac{A_{1\phi} \times 1000}{A_t} = 210.466 \text{ mm}$$

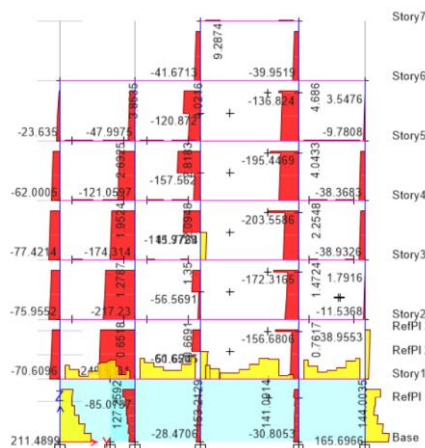
Provide 16 mm ϕ bar @ 175mm c/c in both direction at top and bottom.

Axial Force, Shear Force and Bending Moment Diagram

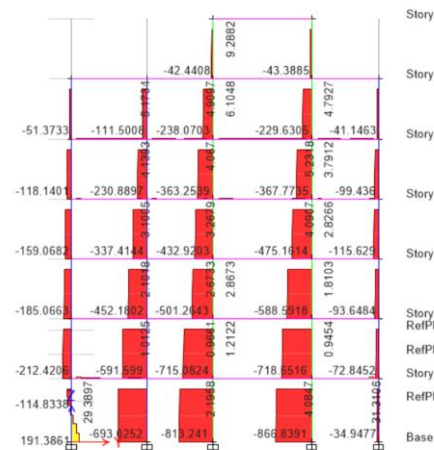
About 2-2 axis

Axial Force in different elevation Diagram

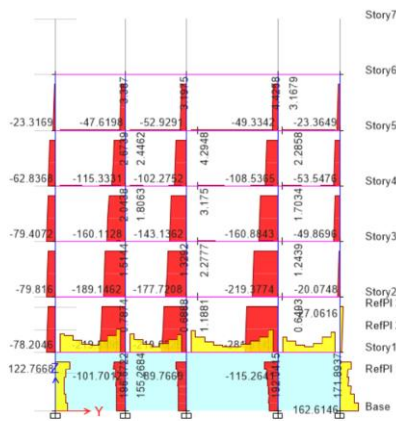
1-1



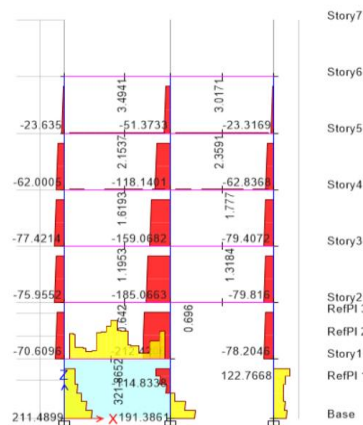
2-2



3-3



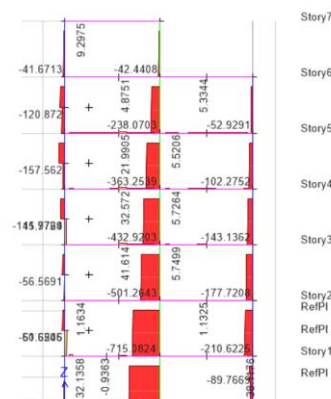
A-A



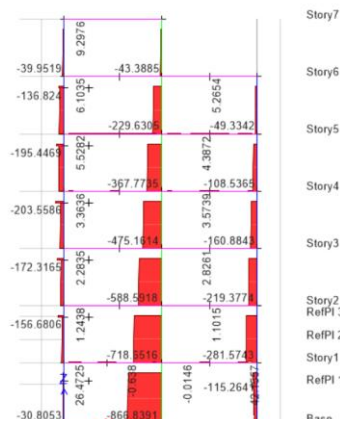
B-B



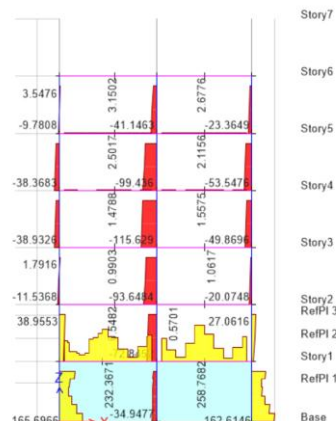
C-C



D-D



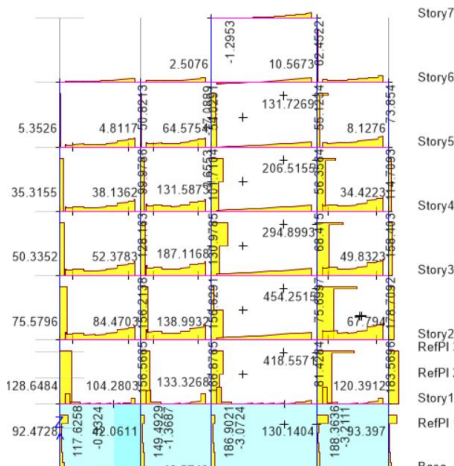
E-E



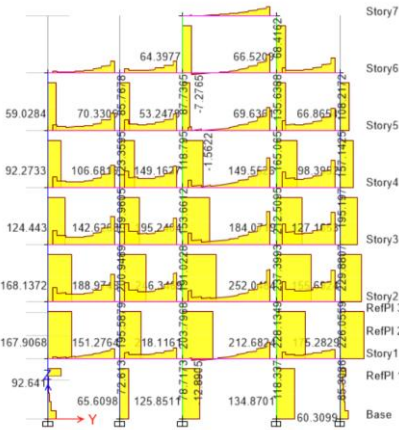
About 2-2 axis

Shear Force in different elevation Diagram

1-1



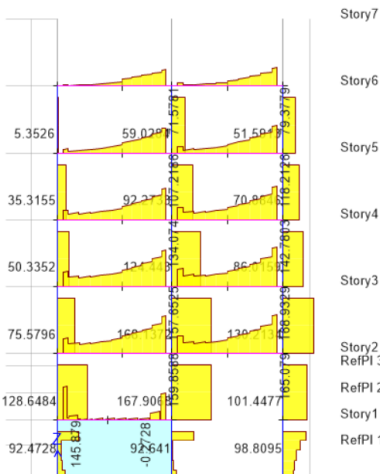
2-2



3-3



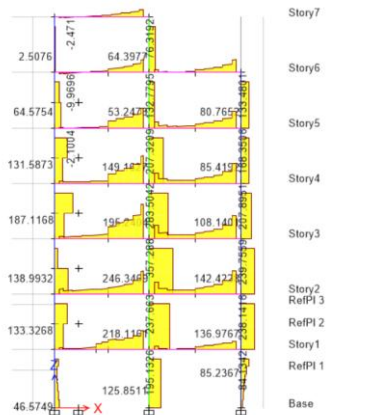
A-A



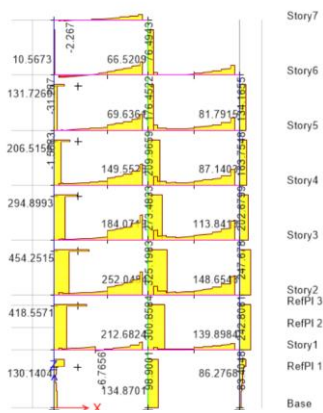
B-B



C-C



D-D

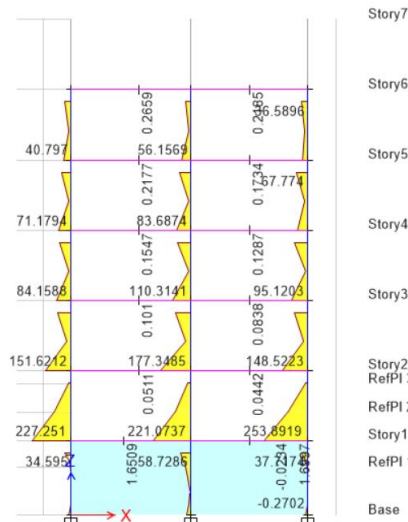


E-E



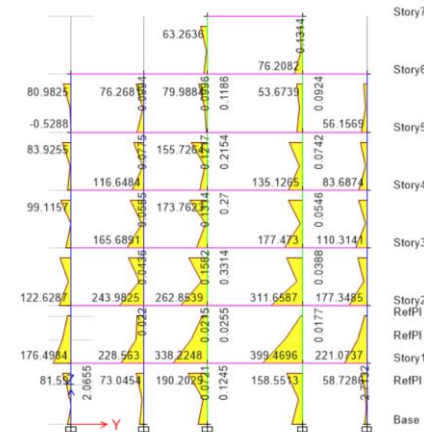
Bending Moment Diagram in different elevation

1-1

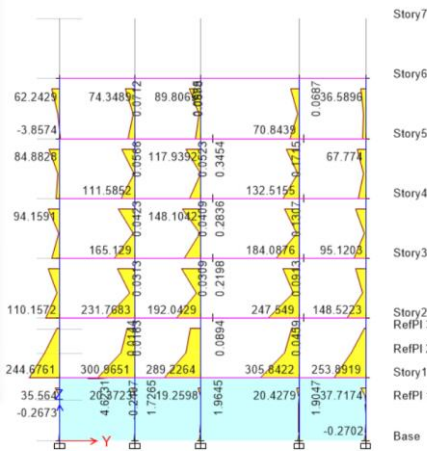


3-3

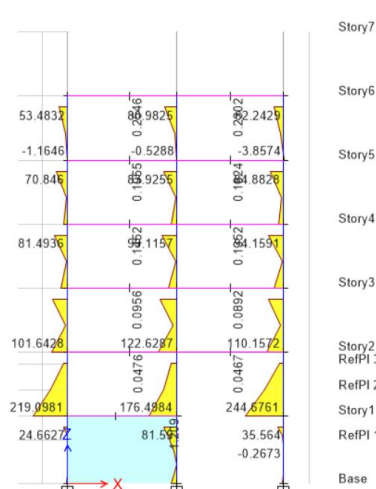
2-2



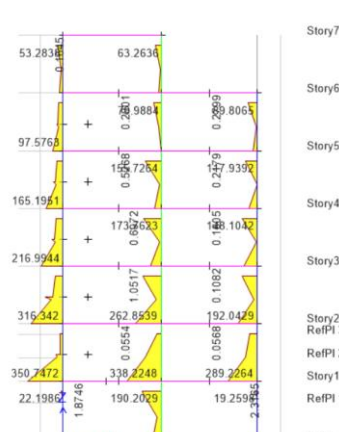
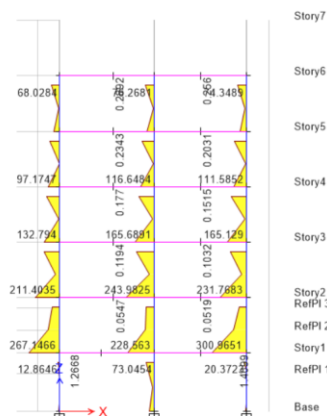
A-A



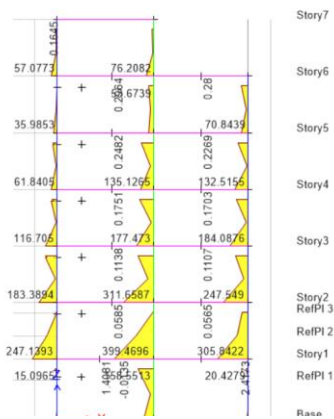
B-B



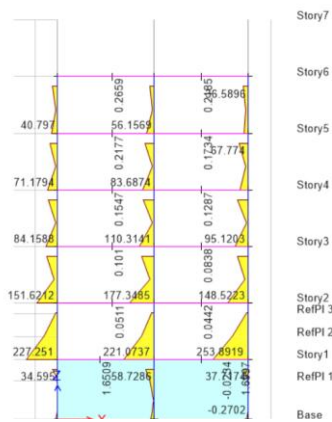
C-C



D-D



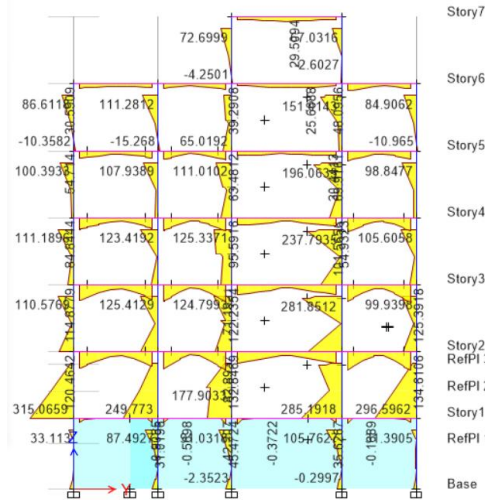
E-E



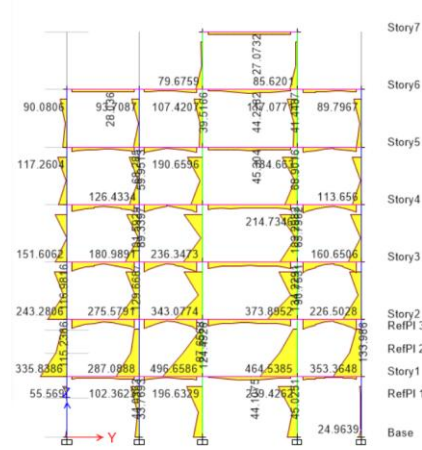
About 3-3 axis

Bending Moment Diagram in different elevation

1-1



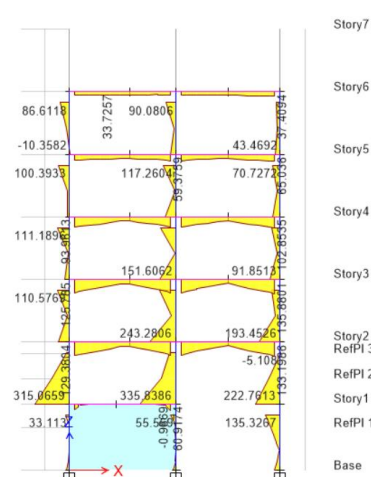
2-2



3-3



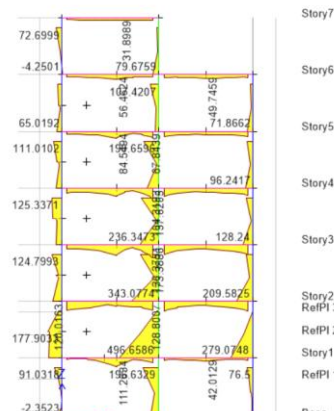
A-A



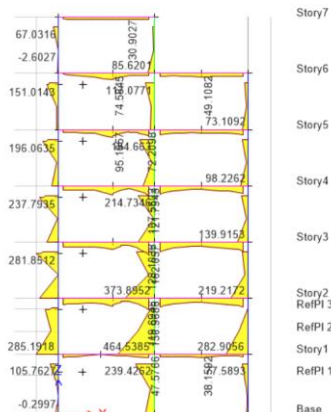
B-B



C-C



D-D



E-E

